

Development of a Deep Reinforcement Learning Agent for Optimal Project Portfolio Budgeting under Uncertain Cash Flows

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Abstract

The path toward a deployable decision-support system for project portfolio management comprises three steps: establishing domain foundations, formulating a tractable decision model, and validating at scale within incumbent systems. The first step is well served by existing literature — contract mechanics, earned value management, S-curve profiles, and portfolio selection models are mature — and the third remains future work. This paper occupies the second step through a relaxation hierarchy applied to a problem absent from the formal literature: *given a fixed amount of capital and a committed portfolio of contracted projects, how should a governing board allocate budget across projects at each discrete period to maximise discounted net cash flow while satisfying hard liquidity constraints and respecting FIDIC contract mechanics??*

Level 1 formulates the problem as a deterministic full-foresight Mixed-Integer Linear Program whose optimum constitutes a provable upper bound no causal policy under uncertainty can exceed. **Level 2** introduces stochastic contractor performance, establishing that the resulting multi-stage stochastic program is intractable at realistic portfolio scales due to exponential scenario-tree growth. **Level 3** recasts the problem as a Constrained Markov Decision Process implemented as a configurable Gymnasium-compatible environment, on which a Proximal Policy Optimisation agent is trained and evaluated against the Level 1 oracle, a two-stage stochastic program, and a rule-based heuristic. The environment — encoding advance payment recovery, milestone payments, retention release, deviation penalties, and a two-condition termination gate with a project-specific cure-period counter — constitutes the primary contribution, positioned as the tractable middle step between established domain foundations and future large-scale deployment.

Keywords: project portfolio management; budget allocation under uncertainty; relaxation hierarchy; constrained Markov decision process; reinforcement learning; FIDIC contracts; earned value management; mixed-integer linear programming.

1 Introduction

1.1 Motivation and Problem Context

Large infrastructure and engineering projects routinely overrun their budgets and miss their schedules. Across a database of thousands of projects spanning seven decades, [Flyvbjerg et al. \(2003\)](#) document that cost overruns are the norm rather than the exception,

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with mean overruns exceeding 20% in transportation infrastructure and reaching 70% in large-scale IT systems. Merrow (2011) reports that fewer than 40% of industrial megaprojects achieve their approved cost and schedule targets. These figures describe individual projects. When a portfolio of such projects is executed simultaneously under a single governing body and a shared capital commitment, the risks compound: a cash-flow shortfall on one project propagates immediately to others, and the decision of how much budget to direct to each project at each period can determine whether the portfolio as a whole succeeds or fails.

Despite this, the formal treatment of the portfolio-level budget allocation problem as a sequential decision-making problem is conspicuously absent from the project portfolio management literature. Research attention has been concentrated at the *selection* stage — which projects to include in the portfolio — while the *operational governance* stage — how to manage the portfolio of committed projects through to completion — has received comparatively little analytical treatment (Archer and Ghasemzadeh, 1999; Cooper et al., 2001; Turner, 2014). The present work fills this gap.

We consider a portfolio of n projects, each governed by a bilateral contract consistent with FIDIC forms (Fédération Internationale des Ingénieurs-Conseils, 2017), executing concurrently under a fixed capital commitment. A portfolio manager — a board, investment committee, or programme director — exercises a single recurring decision at each discrete planning period: how much budget to allocate to each active project. All strategic and operational outcomes of interest flow through this allocation decision. The formal problem is to find the allocation policy that maximises discounted net cash flow across the portfolio horizon while satisfying hard liquidity constraints, respecting contract mechanics.

1.2 The Project Management Decision Hierarchy

To situate the present contribution precisely, it is useful to distinguish the three nested decision layers that constitute project portfolio management (Project Management Institute, 2017).

Portfolio selection. The selection problem asks: given a candidate set of projects and a capital budget, which subset should the organisation commit to? This is a combinatorial optimisation problem studied extensively in the literature using mean-variance portfolio theory (Markowitz, 1952; Eilat et al., 2006), multi-criteria decision analysis (Martinsuo and Lehtonen, 2007), and metaheuristic methods (Zhang et al., 2020). The present work treats selection as an exogenous input: the portfolio of committed projects is fixed at the start of the planning horizon, and contracts have already been signed. This decomposition is consistent with the two-stage governance model of Turner (2014) and the PMI Standard for Portfolio Management (Project Management Institute, 2017).

Portfolio governance. Once a portfolio is committed, the governing body must monitor and control execution. This is the layer at which the present work operates. The governing body does not manage individual projects directly — that is the contractor’s role — but it controls the budget directed to each project at each period. This single continuous allocation decision is the sole instrument of the portfolio manager. Termination is not a discrete agent action but an endogenous environment mechanic demonstrating project failure; it is preceded by a cure period modelled as a project-specific tolerance counter observable in the portfolio state.

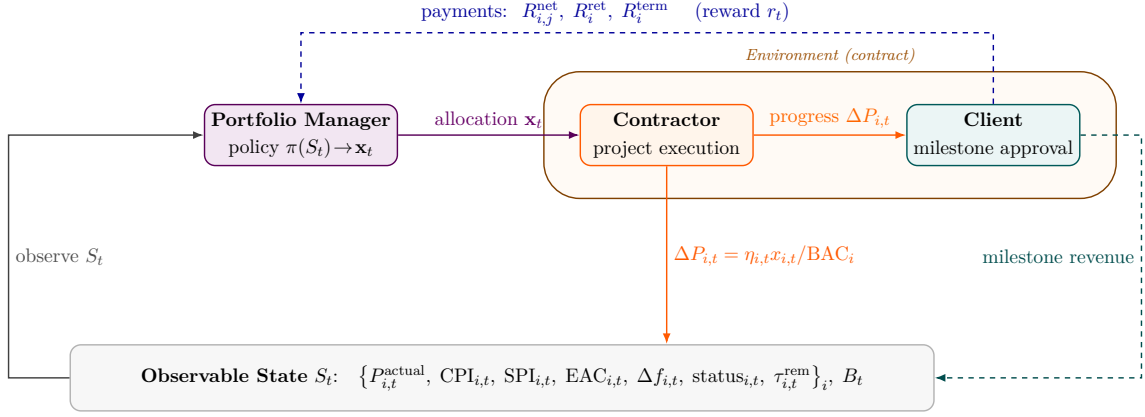


Figure 1: Portfolio governance as a reinforcement learning feedback loop.

Project execution. At the project level, the contractor controls workforce, materials, equipment, and subcontracting. The portfolio manager observes the outcomes of these decisions — cumulative progress, earned value, payment certifications — but does not control them. The efficiency with which a budget allocation is converted into physical progress is determined at the project level and is uncertain from the portfolio manager’s perspective.

This hierarchy is central to the modelling architecture: the formulation developed in Section ?? lives entirely at the portfolio governance layer, with project execution dynamics appearing only as parameters (deterministic at Level 1, stochastic at Levels 2 and 3).

1.3 Contract Mechanics and Cash-Flow Structure

The financial interaction between the portfolio manager (as client) and each contractor is governed by the contract. This paper parameterises all contract mechanics consistently with FIDIC Red, Yellow, and Silver Book forms (Fédération Internationale des Ingénieurs-Conseils, 2017), which represent the dominant international framework for EPC contracting. Understanding the cash-flow structure is essential to understanding the optimisation problem; this subsection provides that foundation.

Milestone-based payment. Rather than paying contractors on a time-basis, FIDIC contracts structure payment around certified milestones: discrete progress thresholds at which the client verifies that a specified scope fraction has been physically completed and releases the corresponding payment. The gross payment at milestone j is a contractually fixed fraction $\phi_{i,j}$ of the contract price CP_i . The contract price relates to the budget at completion through the contractor’s profit margin π_i as $CP_i = (1 + \pi_i) \cdot BAC_i$, and it serves exclusively as the payment accumulation baseline. All EVM performance signals, progress calculations, and termination conditions are normalised against BAC_i . Because payment is contingent on both certified progress and the contractual payment window, the timing of cash inflows is a function of the contractor’s actual performance and the portfolio manager’s allocation decisions, subject to the floor imposed by the contractual earliest certification dates $e_{i,j}$. This creates the fundamental linkage between allocation decisions and cash inflows that makes the problem non-trivial. A portfolio manager operating in a competitive market with a pipeline of future contracts has a direct incentive to complete projects early — not merely on time — because the profit margin π_i is fixed at signing and the only variable the manager controls is how quickly that margin is realised and capital

is freed for redeployment. Earlier completion under fixed margin is equivalent to a higher annualised return on the capital committed (?).

Advance payment and progressive recovery. At project initiation, the client disburses an advance payment $A_i = \alpha_i \cdot CP_i$ to the contractor. This is a timing instrument: it reduces the contractor’s early working-capital burden and is recovered progressively through percentage deductions from subsequent interim payment certificates, with the deduction rate δ_i^{rec} and commencement threshold ψ_i specified as project-specific contract parameters (Fédération Internationale des Ingénieurs-Conseils, 2017). The advance payment creates an immediate cash inflow at project start and a series of reduced inflows at each certified milestone, both of which must be tracked in the portfolio cash balance. Interim milestone payments are additionally subject to stochastic timing delays between certification and receipt, consistent with observed payment practices in international EPC contracting (Aibinu and Jagboro, 2002; Odeh and Battaineh, 2002); the amount transferred equals the contractually fixed net milestone value, but the period of receipt is a random variable.

Retention. A fraction ρ_i of each milestone payment is withheld by the client as a performance bond. The accumulated retention $R_i^{\text{ret}} = \rho_i \cdot CP_i$ is released in full upon project completion. Retention creates a structural completion incentive embedded in the payment mechanics: the larger the retention fraction, the more strongly the cash flow profile rewards completion over partial delivery. In the formulation, retention release enters the portfolio cash balance as a lump inflow in the period the final milestone is certified.

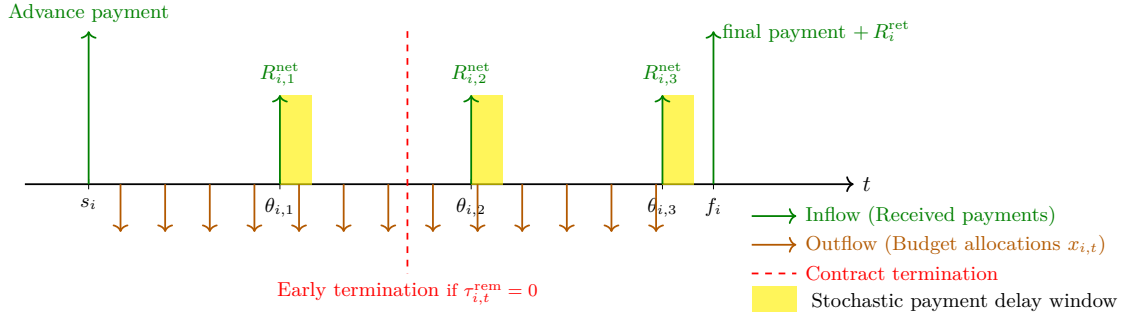


Figure 2: Cash-flow timeline for project i under FIDIC contract mechanics.

Earned value management as a performance signal. The Earned Value Management framework (Fleming and Koppelman, 2010) provides the diagnostic vocabulary used to monitor contractor performance. The Cost Performance Index $CPI_{i,t} = BCWP_{i,t}/ACWP_{i,t}$ measures how much budgeted value has been earned per unit of actual expenditure; values below unity indicate cost overrun. The Schedule Performance Index $SPI_{i,t} = BCWP_{i,t}/BCWS_{i,t}$ measures actual progress relative to planned progress; values below unity indicate schedule lag (Fleming and Koppelman, 2010). The Estimate at Completion $EAC_{i,t} = ACWP_{i,t} + (BAC_i - BCWP_{i,t})/CPI_{i,t}$ projects the total cost of the project at the current cost performance rate, directly analogous to the forecast finish date computation; ? established that this formulation produces stable and accurate cost-at-completion forecasts beyond the 20% completion mark (?). In the present formulation, all three indices are observable state variables; CPI and SPI additionally feed the two forward-looking termination conditions, making their stochastic dynamics — calibrated

from empirical EVM datasets (Acebes et al., 2014; Batselier and Vanhoucke, 2017) — directly consequential for termination timing.

S-curve expenditure profiles. The planned expenditure of a construction or engineering project follows a characteristic sigmoid profile over its planned duration: slow ramp-up in the mobilisation phase, rapid expenditure in the peak execution phase, and tapering in the close-out phase (Kenley and Wilson, 2009). This profile is modelled using the regularized incomplete Beta function $F_i(\tau) = I_\tau(a_i, b_i)$, whose shape parameters (a_i, b_i) are calibrated from project data. The planned cumulative expenditure fraction at normalised time $\tau_{i,t}$ provides the baseline against which actual progress is measured throughout the formulation.

Termination. FIDIC contracts provide for client-initiated termination when the contractor has materially breached the contract — most commonly through sustained failure to meet schedule obligations. Termination triggers a settlement payment based on the value of work completed at the termination date, net of any outstanding advance recovery obligation. In the present model, termination is an endogenous environment mechanic, not a deliberate agent action. It is triggered when two conditions hold simultaneously for a project-specific number of consecutive periods — the termination tolerance τ_i^{tol} . Both conditions are forward-looking trajectory estimates symmetric in structure to one another: Condition 1 fires when the forecast finish date $\hat{f}_{i,t}$, extrapolated from the current schedule performance rate, exceeds the planned finish by more than the contractual schedule tolerance; Condition 2 fires when the Estimate at Completion $EAC_{i,t} = ACWP_{i,t} + (BAC_i - BCWP_{i,t})/CPI_{i,t}$ exceeds the project-specific maximum tolerable cost overrun $(1 + \mu_i) \cdot BAC_i$. This joint persistence requirement models the FIDIC cure period (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008), the contractually prescribed interval during which both parties are obliged to attempt performance recovery before the client may formally terminate. the tolerance counter $\tau_{i,t}^{\text{rem}}$ — periods remaining before termination executes — is part of the observable environment state; the agent cannot override its dynamics but observes its value and can influence it indirectly through allocation decisions that drive $SPI_{i,t}$ and $CPI_{i,t}$. When the counter reaches zero, termination executes in the following period, and its timing affects the portfolio cash balance through the settlement payment R_i^{term} , which may be positive or negative depending on the advance recovery position. The settlement formula computes value as the fraction of work physically delivered multiplied by the contract price — this is not a penalty construct but a direct valuation of transferred productive assets, consistent with how FIDIC termination clauses treat partially completed works as resources whose ownership and value must be accounted for at contract close (Fédération Internationale des Ingénieurs-Conseils, 2017). This grounding means the model’s financial signals reflect real asset values, not arbitrary penalty weights.

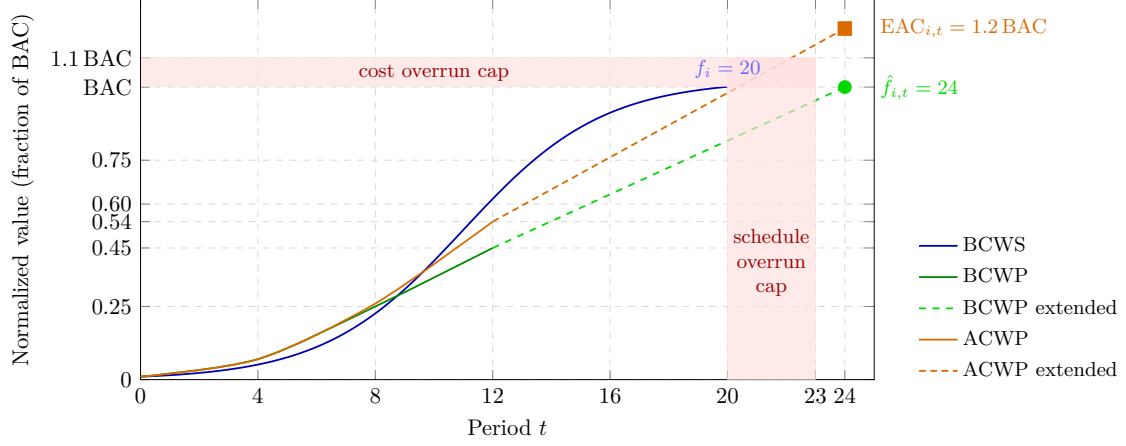


Figure 3: Illustration of EVM-based project termination conditions.

1.4 The Three-Level Modelling Arc

The formulation of Section ?? proceeds across three levels of increasing generality. This subsection provides the conceptual roadmap; each level is developed formally in its corresponding subsection.

Level 1: Deterministic full-foresight MILP. The natural starting point for any optimisation problem is its deterministic version: assume all uncertain parameters are known, and find the optimal decision sequence. In the portfolio budget allocation problem, the key uncertain parameter is the budget-to-progress efficiency $\eta_{i,t}$ — the scalar that determines how much physical progress a given budget allocation yields in each period. If $\eta_{i,t}$ is treated as a known constant $\bar{\eta}_{i,t}$, the progress recursion becomes a deterministic linear map from budget allocations to physical outcomes, and the entire optimisation problem collapses to a Mixed-Integer Linear Program (MILP).

The MILP is not proposed as a practical solution method — no real portfolio manager has advance knowledge of contractor efficiency trajectories. Rather, it serves two purposes. First, its optimal value Z_{L1}^* is a provable upper bound on the performance of any causal policy operating under uncertainty: since the MILP optimises with perfect foresight, it can do no worse than any policy that must make decisions without knowing future outcomes. Second, it provides an LP foundation from which the stochastic extensions of Levels 2 and 3 are constructed by systematic relaxation of the deterministic assumption.

The MILP requires two structural choices that are standard in combinatorial optimisation. Milestone certification is a binary event — a milestone is either certified or not — captured by introducing binary decision variables $u_{i,j,t} \in \{0, 1\}$. The objective is pure discounted net cash flow: inflows minus outflows, discounted at rate γ . No deviation penalty enters the objective; the deviation state variable $\Delta f_{i,t}$ is tracked solely to feed the termination conditions, whose consequences — loss of all future milestone payments, retention, and a potentially negative settlement — provide sufficient incentive for the agent to manage schedule performance without requiring an additional penalty term (Ng et al., 1999). The absence of a penalty weight is not merely a modelling simplification but reflects the actual structure of FIDIC contracts, which do not penalise schedule deviation directly but impose consequences through the termination mechanism. A portfolio manager who internalises this structure will front-load allocations under discounting not because a penalty pushes them to do so, but because earlier milestone certification yields

higher discounted inflows — a pure income-maximisation motive requiring no additional calibration parameters. A second structural feature is the termination mechanic: under deterministic $\bar{\eta}_{i,t}$, both termination conditions and the tolerance counter trajectory $\tau_{i,t}^{\text{rem}}$ are fully pre-computable, so termination periods t_i^{term} and their settlements R_i^{term} enter the MILP as fixed parameters rather than optimisation constraints — a simplification that does not carry over to Levels 2 and 3, where termination timing is endogenous to the stochastic trajectory.

Level 2: Stochastic extension and intractability. When $\eta_{i,t}$ is replaced by a random variable — as it must be to reflect reality — the MILP becomes a multi-stage stochastic program (Birge and Louveaux, 2011). The scenario-tree representation of this program grows exponentially: if $\eta_{i,t}$ takes K discrete values at each of H periods for each of n projects, the scenario tree has K^{nH} leaf nodes. For a portfolio of ten projects over a 24-month horizon with five efficiency scenarios per project per period, this is 5^{240} leaf nodes — a number that renders exact stochastic programming computationally intractable. This is not merely a computational inconvenience; it is the fundamental barrier that motivates the Level 3 reformulation.

Level 3: Constrained MDP and RL agent. The Constrained Markov Decision Process (Altman, 1999) provides the tractable reformulation. Rather than optimising over all scenario-tree paths simultaneously, the CMDP formalises the problem as a sequential interaction between a policy and an environment: at each period, the policy observes the current portfolio state, selects a budget allocation, and the environment transitions stochastically to the next state. The key insight is that the CMDP does not require explicit scenario enumeration: the policy is optimised through repeated interaction with the environment, using sampled trajectories rather than exhaustive scenario trees.

The mapping from MILP to CMDP is exact in structure. The decision variables $\{x_{i,t}\}$ become the action space; the state variables

$$(P_{i,t}^{\text{actual}}, \text{ACWP}_{i,t}, \text{CPI}_{i,t}, \text{SPI}_{i,t}, \text{EAC}_{i,t}, \Delta f_{i,t}, \text{status}_{i,t}, \tau_{i,t}^{\text{rem}}, B_t)$$

become the observation space; and the objective function — discounted net cash flow becomes the reward signal. The CMDP is implemented as a configurable Gymnasium-compatible environment (Towers et al., 2024), and a Proximal Policy Optimisation agent (Schulman et al., 2017) is trained on it using domain randomisation (Tobin et al., 2017) to ensure generalisation across project types and contract structures.

1.5 The Operations Research Lineage

The present work is situated squarely within the Industrial Engineering and Operations Research tradition. The mathematical tools employed — mixed-integer linear programming, stochastic programming, dynamic programming, and Markov decision processes — share a common intellectual lineage that traces directly to Dantzig’s simplex method (Dantzig, 1963), Bellman’s principle of optimality (Bellman, 1957), and the stochastic programming framework of Dantzig (1955).

The MDP formalisation used in Level 3 is, mathematically, a computationally tractable approach to solving a Bellman equation in a high-dimensional state space. The state space in the present problem — $9n + 1$ continuous and categorical variables for a portfolio of n projects — is too large for tabular dynamic programming and too structured for model-free

value-iteration methods, but well-suited to policy-gradient methods that learn a parameterised policy directly. PPO (Schulman et al., 2017) is the policy-gradient algorithm of choice for its stability properties in continuous action spaces and its established use in operations management domains (Giannoccaro and Pontrandolfo, 2002; Peng and Huang, 2021).

The reinforcement learning agent is therefore not presented as a departure from the OR tradition, but as its continuation into the regime where the curse of dimensionality makes analytical dynamic programming intractable. The Level 1 MILP and Level 2 stochastic program are not straw men; they are the canonical OR solutions to the deterministic and stochastic versions of the problem, respectively, and both serve as baselines against which the Level 3 agent is evaluated.

1.6 Practical Deployment Context

Incumbent project management information systems — most prominently Oracle Primavera P6 and Microsoft Project — provide rich project-level schedule and cost tracking but lack a portfolio-level optimisation layer. The portfolio manager’s recurring budget allocation decision is currently supported, at best, by spreadsheet models, earned value dashboards, and expert judgement. The environment developed in Level 3 constitutes the analytical engine that such a portfolio-level decision-support module would require: it integrates contract mechanics, performance uncertainty, and multi-project cash-flow dynamics and a portfolio-derived planning horizon in a single configurable framework. This positions the contribution not only as a theoretical advance but as a step toward a deployable module for portfolio management information systems, a direction explicitly identified as a priority in the project management software literature (Project Management Institute, 2017).

1.7 Paper Organisation

The remainder of the paper is organised as follows. Section ?? reviews the four literature streams on which the work draws: project management foundations, project portfolio management and selection, PPM budgeting and cash-flow modelling, and reinforcement learning for sequential resource allocation. Section ?? develops the three-level problem formulation: the deterministic MILP (Level 1), the stochastic extension and intractability argument (Level 2), and the CMDP reformulation and Gymnasium-compatible environment (Level 3). Section ?? presents the empirical evaluation: calibration of environment parameters, training protocol, and comparison of the PPO agent against the Level 1 MILP oracle, a two-stage stochastic program, and a rule-based heuristic. Section ?? concludes with implications for portfolio management practice, limitations of the framework, and directions for future research including integration with incumbent PMIS tools.

2 Literature Review

2.1 Overview and Review Strategy

This section maps the intellectual lineage of the present work across four adjacent knowledge domains: project management (PM) foundations, project portfolio management (PPM) and selection, PPM budgeting and cash-flow modelling, and reinforcement learning (RL) applied to sequential resource-allocation problems. Together these streams define both the theoretical foundation and the gap that motivates the proposed framework. The review

follows the PRISMA 2020 reporting guidelines (Page et al., 2021) to ensure transparency and reproducibility in source identification, screening, and inclusion decisions.

2.1.1 Search and Screening Protocol

Literature was retrieved from Google Scholar, Semantic Scholar, and Scopus using a progressively narrowing search strategy. Queries began at the broadest relevant domain (*project management*) and were successively refined through *project portfolio management*, *portfolio budgeting and cash-flow optimisation*, *earned value management under uncertainty*, and *reinforcement learning for project scheduling and resource allocation*. Citation-network traversal — backward and forward snowballing from high-citation anchor papers — was used to recover seminal works not surfaced by keyword search.

Records were screened in three sequential stages consistent with the PRISMA 2020 flow diagram (Appendix A). **Stage 1** (identification) assessed machine readability and bibliographic completeness. **Stage 2** (relevance filtering) applied cosine-similarity ranking against the research hypothesis using sentence-transformer embeddings, retaining sources above a calibrated relevance threshold. **Stage 3** (structured extraction) classified each retained paper by contribution type, key variables, modelling paradigm, and empirical basis. The PRISMA flow diagram quantifying records at each stage is provided in Appendix A.

2.2 Project Management: Foundational Concepts

The academic study of project management draws on schedule control (CPM/PERT), earned value management (EVM), and risk modelling as its core quantitative pillars (Fleming and Koppelman, 2010; Project Management Institute, 2021). EVM introduced the Cost Performance Index (CPI) and Schedule Performance Index (SPI) as real-time diagnostic signals relating budgeted cost of work performed to actual expenditure and planned schedule — providing the performance vocabulary that the present model inherits directly. Values of $CPI < 1$ signal cost overrun; $SPI < 1$ signals schedule lag. Stochastic extensions of EVM (Acebes et al., 2014; Batselier and Vanhoucke, 2017) and Monte Carlo simulation frameworks (Kwak and Ingall, 2007) have modelled performance uncertainty at the project level, establishing that CPI and SPI follow empirically characterisable distributions over project life.

S-curve analysis, formalised by Kenley and Wilson (2009) for construction cash flows, established that budget consumption follows characteristic sigmoid profiles whose shape parameters are project-type- and contract-form-dependent. The regularized incomplete Beta function provides a flexible single-family parameterisation of this profile (Barraza et al., 2004), and is adopted directly in the planned expenditure model of the present formulation.

Large-scale empirical evidence on project performance is provided by Merrow (2011) through the Independent Project Analysis database, and by Flyvbjerg et al. (2003); Flyvbjerg (2017) through infrastructure cost-overrun datasets spanning decades and geographies. These datasets inform both the calibration of uncertainty distributions in the present model and the benchmarks against which synthetic environment parameters are validated. Complementary empirical benchmarks are drawn from AACE International and Construction Industry Institute cost-engineering databases (Christensen, 1993).

Contract governance structures — particularly FIDIC forms (Red, Yellow, and Silver Books) — prescribe the financial interaction between owner and contractor through milestone payment schedules, advance payment and recovery mechanisms, and retention clauses (Fédération Internationale des Ingénieurs-Conseils, 2017). The Estimate at Com-

pletion $EAC_{i,t} = ACWP_{i,t} + (BAC_i - BCWP_{i,t}) / CPI_{i,t}$ extends CPI into a forward-looking cost trajectory signal; ? established across 155 contracts that this formulation produces the most accurate and stable cost-at-completion forecasts, with reliability increasing beyond the 20% completion mark (?). The present model adopts EAC as the second termination condition signal, directly parallel to the forecast finish deviation used in Condition 1. Under FIDIC Sub-Clause 14.2, the advance payment is an interest-free mobilisation loan recovered through percentage deductions from interim payment certificates: if not otherwise specified in the Appendix to Tender, deductions commence once cumulative certified interim payments exceed 10% of the Accepted Contract Amount and are applied at a rate of 25% of each subsequent payment certificate until the advance is fully recovered (Fédération Internationale des Ingénieurs-Conseils, 2017). The present model parameterises this mechanism with a project-specific commencement threshold ψ_i and deduction rate δ_i^{rec} , accommodating the full range of particular conditions encountered in practice. These mechanisms translate directly into the cash inflow and outflow dynamics formulated in Section ??.

2.3 Project Portfolio Management: Selection and Governance

Project portfolio management formalises the problem of allocating an organisation’s finite resources across a competing set of candidate projects to maximise strategic value (Archer and Ghasemzadeh, 1999; Cooper et al., 2001). The literature distinguishes two nested decision layers: *portfolio selection* — the binary or mixed-integer optimisation of which projects to include — and *portfolio governance* — the ongoing operational oversight of approved, active projects.

Portfolio selection models have been extensively developed. Markowitz-style mean-variance frameworks adapted from financial portfolio theory (Markowitz, 1952) have been applied to project baskets to trade off expected NPV against variance (Eilat et al., 2006). Multi-criteria decision analysis extensions incorporate strategic alignment, risk ratings, and resource interdependencies (Martinsuo and Lehtonen, 2007; Killen et al., 2008). Meta-heuristic approaches — genetic algorithms, particle swarm optimisation — address large combinatorial instances (Zhang et al., 2020). Resource-constrained portfolio selection under capital rationing has been studied by Kavadias and Loch (2003), whose no-borrowing constraint is adopted directly in the present formulation.

The present work *explicitly adopts the output of portfolio selection as an exogenous input*. The selection problem is treated as solved upstream; the focus is entirely on the operational governance layer that receives a committed portfolio and must manage it through execution. This decomposition is consistent with the two-stage governance model of Turner (2014) and the PMI Standard for Portfolio Management (Project Management Institute, 2017), which separates strategic selection from operational execution oversight. In doing so, this work builds directly on the selection literature while advancing the frontier into the governance domain that has received substantially less formal analytical treatment.

2.4 PPM Budgeting, Cash-Flow Modelling, and Uncertainty

Within portfolio governance, budget allocation across active projects over time represents a high-frequency, high-stakes operational decision. Despite its practical centrality, this specific problem has received limited formal treatment relative to portfolio selection. Existing work addresses project-level cash-flow modelling (Kenley and Wilson, 2009; Park and Peña-Mora, 2004), EVM-based forecasting of individual project cost-at-completion (Kim and Ballard, 2002; Acebes et al., 2014), and multi-project resource levelling (Lova

et al., 2000; Van Peteghem and Vanhoucke, 2010). However, these contributions treat budgets as deterministic constraints rather than as dynamic decision variables subject to portfolio-level sequential optimisation.

Cash-flow uncertainty in construction and engineering projects arises from contractor performance variability, payment timing delays between certification and receipt, scope changes, and external market shocks (Odeh and Battaineh, 2002; Doloi, 2012; Aibinu and Jagboro, 2002). Empirical studies of payment practices in international construction contracting document systematic delays between milestone certification and client payment, with delay distributions that vary by contract type, jurisdiction, and client financial health (Aibinu and Jagboro, 2002; Assaf et al., 1995); the present model treats certified-milestone payment timing as a stochastic variable while holding the payment amount fixed at the contractually determined net value after advance recovery deductions and retention withholding. Stochastic programming (Birge and Louveaux, 2011) provides the canonical OR framework for optimisation under such uncertainty, and its two-stage variant has been applied to project scheduling under resource uncertainty (Lova et al., 2000). However, multi-stage stochastic programs for portfolio-level sequential budget allocation face scenario-tree explosion as the number of projects and periods increases. The effective planning horizon is itself portfolio-dependent—determined by the latest project finish—compounding the explosion, since H grows with the most delayed project in the committed set. Both effects are formalised in Section 3.5..

The gap in the literature is precise: *no established framework models the portfolio-level budget allocation decision as a sequential stochastic optimisation problem in which a central decision-maker allocates funds across multiple active projects whose cash inflow and outflow trajectories are uncertain, coupled through a shared capital constraint, and subject to contractual earliest certification windows that bound the speed at which progress can be converted into cash inflows.* The present work addresses this gap. The objective is pure discounted net cash flow; no deviation penalty enters the objective function. This is a deliberate design choice: the consequences of sustained deviation — termination and loss of all future milestone payments and retention — are sufficient to make deviation undesirable without a separate penalty term, and including one would introduce an arbitrary scalarisation weight with no natural calibration from contract data (Ng et al., 1999). Symmetrically, the discount factor γ encodes not only time preference but the opportunity cost of tied-up capital in a competitive environment where future projects are foregone while current ones remain incomplete (?). This interpretation gives γ a concrete calibration anchor beyond abstract preference — it can be estimated from the firm’s observed pipeline redeployment rates or from prevailing cost-of-capital benchmarks. The termination settlement further grounds the model in real asset values: the settlement payment computes the liquidation value of assets in place at termination (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001), not an arbitrarily weighted deviation penalty. A consequence is that the learned policy may produce allocation trajectories that concentrate expenditure around milestone certification thresholds, mirroring documented contractor financial strategy in milestone-payment contracts (Kenley and Wilson, 2009; Kaka and Price, 1993; Arditi and Chotibhongs, 2005).

2.5 Contractor Underperformance, Cure Periods, and Termination

When a contractor fails to meet schedule or cost performance thresholds, international EPC contracts prescribe a structured escalation sequence before termination may be exercised. Under FIDIC Sub-Clause 15.1, the client must issue a formal notice of default and allow a cure period — typically 28 days — during which the contractor is obliged to

demonstrate measurable recovery before the client may proceed to termination (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008). This cure period is not a discretionary grace; it is a contractual obligation whose omission constitutes wrongful termination. Empirical studies of international EPC disputes confirm that failure to observe the cure period is among the most frequently litigated grounds in contractor termination claims (Ndekugri and Rycroft, 2008; Ennis and Burr, 2021).

The behavioural project management literature has studied the conditions under which organisations fail to terminate underperforming projects at the economically rational point. Escalation of commitment (Staw and Ross, 1987) — the tendency to continue investing in failing projects beyond what expected value justifies — has been documented across construction, IT, and infrastructure programmes. Real options theory (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001) provides the normative counterpart: termination is the exercise of an abandonment option whose value exceeds continuation NPV once performance has deteriorated past a recoverable threshold. Together these streams establish that the decision to persist with or terminate an underperforming project is neither purely contractual nor purely financial — it involves performance signals, contractual obligations, and timing all at once.

The present model integrates all three dimensions into a single environment mechanic. Termination is triggered when two conditions hold simultaneously: the forecast finish deviation has exceeded the contractual schedule tolerance (Condition 1), and the Estimate at Completion $EAC_{i,t} = ACWP_{i,t} + (BAC_i - BCWP_{i,t})/CPI_{i,t}$ has exceeded the project-specific maximum tolerable cost overrun $(1 + \mu_i) \cdot BAC_i$ (Condition 2). Both conditions are forward-looking trajectory estimates symmetric in structure: remaining planned work divided by current performance rate, projected forward to completion (Fleming and Koppelman, 2010; ?). Rather than executing immediately, the environment decrements a project-specific tolerance counter $\tau_{i,t}^{\text{rem}}$, which represents the remaining cure period. Termination executes only when the counter reaches zero; if either condition lapses before then, the counter resets to its project-specific maximum τ_i^{tol} . This mechanic is grounded simultaneously in FIDIC contract law (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008), the real options framing of the abandonment threshold (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001), the EVM literature’s validation of EAC as a stable forward cost signal (??), and the escalation-of-commitment literature’s evidence that observable performance signals must be sustained — not transient — before termination is justified (Staw and Ross, 1987). The tolerance counter is part of the observable portfolio state, giving the RL agent an anticipatory signal that reflects contractual reality without granting it discretionary control over the termination outcome. From the portfolio manager’s perspective, this anticipatory signal serves a dual function — it flags projects approaching contractual failure while also signalling the point at which continued allocation yields diminishing returns relative to the settlement value of work already delivered. A rational manager reading a counter approaching zero in a high-opportunity-cost environment would recognise that the real value of future milestone payments is eroding while the settlement value of work in place is fixed in nominal terms — creating a natural reallocation incentive that the model captures through the interaction of γ and R_i^{term} without requiring explicit inflation modelling (Dixit and Pindyck, 1994; Fédération Internationale des Ingénieurs-Conseils, 2017).

2.6 Reinforcement Learning for Sequential Resource Allocation

Reinforcement learning provides a principled framework for sequential decision-making under uncertainty (Sutton and Barto, 2018). The MDP formalisation — state, action,

transition, reward — maps naturally onto the portfolio budget allocation problem, and the constrained MDP extension (Altman, 1999) accommodates the hard budget feasibility constraints that the portfolio manager must satisfy at every period.

RL has been applied to project scheduling (Peng and Huang, 2021; Liu et al., 2023), resource-constrained project scheduling problems (RCPSP) (Tassel et al., 2023), and supply chain inventory management (Giannoccaro and Pontrandolfo, 2002). In each domain, RL demonstrates advantages over classical OR methods when state spaces are large, transition dynamics are stochastic, the problem is non-stationary over the decision horizon, and the horizon length is itself uncertain at the outset of planning. These are precisely the characteristics of the portfolio budget allocation problem at Levels 2 and 3.

Proximal Policy Optimisation (Schulman et al., 2017) is adopted as the solution algorithm for its stability in continuous action spaces, its sample efficiency relative to earlier policy-gradient methods, and its established use in operations management domains. Domain randomisation (Tobin et al., 2017) — randomising environment parameters during training to improve out-of-sample policy robustness — is adopted to ensure the learned policy generalises across the range of project types, contract structures, and market conditions represented in the calibration data. Walk-forward validation (Cerqueira et al., 2020) is used as the empirical evaluation protocol to guard against data leakage given the time-series structure of project cash-flow data.

Direct application of RL to portfolio-level budget allocation remains absent from the literature. The present work provides the first formulation of this problem as a CMDP and the first empirical demonstration that a PPO agent can learn non-trivial allocation policies in a calibrated synthetic environment.

2.7 Synthesis: Positioning the Present Contribution

Table 1 maps representative works across the four literature streams against the key modelling dimensions of the present framework. Three structural gaps emerge.

First, the transition from portfolio *selection* to portfolio *operational governance* has not been modelled as a sequential decision problem. Selection literature assumes away post-selection management; project scheduling literature ignores portfolio interdependencies and the shared capital constraint.

Second, existing stochastic models of project cash-flow uncertainty operate at the individual project level and do not feed uncertainty upward into a portfolio-level decision architecture with a shared budget constraint.

Third, the endogenous mechanics of project failure — the joint modelling of forecast schedule deviation, EAC-based cost overrun trajectory, a contractual cure period, and termination settlement — have not been embedded within a portfolio-level sequential decision framework in which budget allocation is the sole control variable.

These gaps are not isolated oversights; they reflect an incomplete progression along the path toward a deployable portfolio-level decision-support system. Figure 4 situates this progression as three sequential steps. **Step 1** — domain grounding — is well established by the literature reviewed above: contract mechanics, EVM performance signals, S-curve expenditure profiles, and portfolio selection models collectively provide the inputs and boundary conditions that any operational governance model requires. **Step 2** — tractable problem formulation — is the gap. Bridging it requires a relaxation hierarchy: the governance problem must first be stated precisely under idealised assumptions (a deterministic MILP establishing a performance ceiling), then shown to be intractable under realistic stochastic assumptions (exponential scenario-tree growth), and finally reformulated into a tractable approximation (a CMDP solved by a policy-gradient agent).

No existing work completes this hierarchy at the portfolio level. **Step 3** — validated deployment within incumbent project management information systems such as Oracle Primavera P6 — remains future work, contingent on the tractable formulation that Step 2 must first provide.

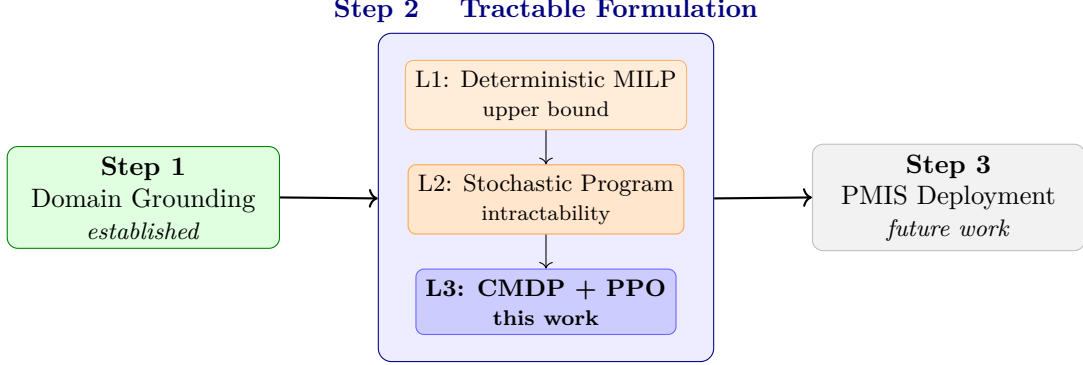


Figure 4: Solution maturity progression toward deployable portfolio-level budget allocation.

The present work addresses all three gaps and completes Step 2 of this progression through the relaxation hierarchy described above and developed formally in Section ?? . The Gymnasium-compatible CMDP environment — with parameterisable contract mechanics, stochastic performance model and termination — constitutes the primary methodological contribution. The PPO agent demonstrates its utility and provides the basis for empirical evaluation against classical OR baselines in Section ?? .

Table 1: Representative literature mapped against modelling dimensions of the present framework. *Decision type*: S = selection, Sc = scheduling, B = budgeting. ✓ = fully addressed; ◦ = partially addressed; – = not addressed.

Work	Portfolio level	Decision type	Sequential decisions	Stochastic cash flows	Contract mechanics	Termination settlement
Eilat et al. (2006)	✓	S	–	–	–	–
Klerides and Hadjiconstantinou (2019)	✓	S	–	◦	–	–
Huchzermeier and Loch (2001)	◦	S	✓	✓	–	✓
Kenley and Wilson (2009)	–	B	–	◦	◦	–
Acebes et al. (2014)	–	B	◦	✓	–	–
Lova et al. (2000)	✓	Sc	–	–	–	–
Peng and Huang (2021)	–	Sc	✓	◦	–	–
Tassel et al. (2023)	–	Sc	✓	–	–	–
Present work	✓	B	✓	✓	✓	✓

3 Problem Formulation

The formulation proceeds across three levels of increasing generality, all sharing the environment primitives defined in this section. **Level 1** (Section 3.4) establishes the deterministic full-foresight MILP whose optimum is a provable upper bound on any causal policy. **Level 2** (Section 3.5) introduces stochastic contractor efficiency and characterises the resulting multi-stage stochastic program, establishing exponential scenario-tree growth.

Level 3 (Section 3.6) reformulates the intractable stochastic program as a Constrained Markov Decision Process, yielding the Gymnasium-compatible environment on which the PPO agent is trained. The three levels form a coherent arc — Level 1 defines the performance ceiling, Level 2 explains why it is unreachable by causal policies, Level 3 provides the tractable path.

3.1 Problem Statement and Scope

The problem is operational, not strategic. A portfolio of n contracted projects is fixed: commitments are signed, start dates are scheduled, and contracts are parameterised consistently with FIDIC Red, Yellow, and Silver Book forms (Fédération Internationale des Ingénieurs-Conseils, 2017). The decision-maker — a portfolio manager, board, or programme director — exercises a single recurring lever at each discrete planning period $t \in \mathcal{T} = \{1, \dots, H\}$: the budget $x_{i,t} \geq 0$ directed to each active project i . The planning horizon H is not a fixed configuration parameter but is determined endogenously as $H = \max_i f_i - \min_i s_i + 1$, where s_i and f_i are the planned start and finish of project i ; the index is shifted so that period $t = 1$ aligns with the earliest project start. All outcomes of interest — milestone achievement, cash inflows, schedule adherence, and project termination — flow through this allocation decision. Portfolio selection is treated as an exogenous upstream input (Archer and Ghasemzadeh, 1999).

Four structural assumptions underlie the entire formulation. (i) *Independent bilateral contracts*: projects interact only through the shared cash balance; no resource or workforce sharing occurs, consistent with standard EPC contracting practice. (ii) *Budget as the sole control*: the contractor’s internal resource decisions are absorbed into a stochastic efficiency parameter $\eta_{i,t}$; the portfolio manager observes how much physical progress a budget allocation yields, not how it is achieved. The timing of that allocation — specifically how quickly milestones are certified — determines the discounted value of inflows under $\gamma < 1$, making the scheduling of spend, not just its total magnitude, the operative decision variable for a return-maximising portfolio manager (?). (iii) *Discrete monthly periods*: $\mathcal{T} = \{1, \dots, H\}$, where H is determined dynamically from the portfolio (see Section 3.2.1), reflects the standard monthly governance cycle of portfolio management (iv) *FIDIC contract mechanics*: milestone payments, advance payment and recovery, retention, and termination settlement are all calibrated parameters, not fixed constants, accommodating the full range of particular conditions encountered in practice.

3.2 Environment Definition

3.2.1 Sets, Indices, and the Decision Variable

$\mathcal{P}$	Projects $\{1, \dots, n\}$, indexed by i ; all committed at horizon start.
$\mathcal{T}$	Planning periods $\{1, \dots, H\}$, indexed by t (calendar months), where $H = \max_{i \in \mathcal{P}} f_i - \min_{i \in \mathcal{P}} s_i + 1$ is derived endogenously from the committed portfolio; period $t = 1$ corresponds to the calendar month of the earliest project start $\min_i s_i$.
$\mathcal{M}_i$	Ordered milestones $\{1, \dots, M_i\}$ for project i , indexed by j , with progress thresholds $\theta_{i,1} < \dots < \theta_{i,M_i} = 1$.
$\mathcal{A}_i^{\text{plan}}$	Planned execution window $\{s_i, \dots, f_i\}$; anchors S-curve computations.
$\mathcal{P}_t^{\text{active}}$	Projects active at period t ; fixed at Level 1, endogenous at Levels 2–3.

$\mathcal{M}_i^{\text{cert}}(t)$	Milestones of project i certified by period t : $\{j : \theta_{i,j} \leq P_{i,t}^{\text{actual}} \text{ and } t \geq e_{i,j}\}$, where $e_{i,j} \in \mathcal{T}$ is the contractually specified earliest certification period for milestone j .
$x_{i,t} \geq 0$	Decision variable: budget allocated to project i at period t . The period- t action vector is $\mathbf{x}_t = (x_{i,t})_{i \in \mathcal{P}_t^{\text{active}}}$.
$B_t \geq 0$	Portfolio cash balance at the start of period t ; B_1 is given.

3.2.2 Progress and Expenditure Dynamics

Budget $x_{i,t}$ converts to physical progress through the efficiency scalar $\eta_{i,t}$, which absorbs all sources of conversion uncertainty (contractor productivity, supply chain, weather):

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \eta_{i,t} \cdot \frac{x_{i,t}}{\text{BAC}_i}, \quad \text{ACWP}_{i,t} = \text{ACWP}_{i,t-1} + x_{i,t}, \quad (1)$$

with $P_{i,0}^{\text{actual}} = \text{ACWP}_{i,0} = 0$. Here BAC_i is the Budget at Completion — the total planned contractor expenditure for project i , which serves as the EVM cost basis and the normalisation denominator for all progress and performance calculations. At Level 1, $\eta_{i,t}$ is replaced by a known deterministic value $\bar{\eta}_{i,t}$; at Levels 2–3 it is a stochastic primitive realised only after $x_{i,t}$ is committed.

The planned cumulative expenditure fraction follows a normalised S-curve (Kenley and Wilson, 2009; Barraza et al., 2004) parameterised by shape parameters (a_i, b_i) :

$$P_{i,t}^{\text{plan}} = I_{\tau_{i,t}^c}(a_i, b_i), \quad \tau_{i,t} = \frac{t - s_i + 1}{D_i^{\text{plan}}}, \quad \tau_{i,t}^c = \min\{1, \max\{0, \tau_{i,t}\}\}, \quad (2)$$

where $I_\tau(\cdot, \cdot)$ is the regularized incomplete Beta function and $D_i^{\text{plan}} = f_i - s_i + 1$ is the planned duration. This profile is a fixed parameter derived at commitment; it is never a decision variable.

3.2.3 Contractual Cash-Flow Mechanics

Payment inflows follow FIDIC mechanics (Fédération Internationale des Ingénieurs-Conseils, 2017). At project initiation ($t = s_i$), the portfolio manager receives the advance payment $A_i = \alpha_i \cdot CP_i$, where α_i is the advance ratio and $CP_i = \text{BAC}_i(1 + \pi_i)$ is the contract price with profit margin π_i . The contract price is the sole payment accumulation baseline: all milestone revenues, advance payment, retention, and final settlement sum to CP_i . All EVM performance signals and progress calculations are normalised against BAC_i .

Milestone j is certified at the first period t at which $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$ and $t \geq e_{i,j}$, where $e_{i,j}$ is the contractually specified earliest certification period. Under FIDIC Sub-Clause 14.3, interim payment applications may not be submitted before the intervals stated in the Appendix to Tender (Fédération Internationale des Ingénieurs-Conseils, 2017); $e_{i,j}$ parameterises this constraint directly, preventing a portfolio manager from accelerating cash inflows beyond the contractual payment schedule regardless of how quickly physical progress is achieved. The net payment received, after retention withholding at rate ρ_i and advance recovery deduction at rate δ_i^{rec} (commencing once cumulative certified payments exceed threshold ψ_i), is:

$$R_{i,j}^{\text{net}} = \phi_{i,j} \cdot CP_i(1 - \rho_i) - \delta_i^{\text{rec}} \cdot \phi_{i,j} \cdot CP_i \cdot \mathbf{1}\left\{\sum_{k \leq j} \phi_{i,k} \geq \psi_i\right\}, \quad (3)$$

where $\phi_{i,j}$ is the contractual weight of milestone j with $\sum_j \phi_{i,j} = 1$. This parameterisation is consistent with FIDIC Sub-Clause 14.2, under which advance recovery deductions commence once cumulative certified interim payments exceed a specified fraction of the contract price (Fédération Internationale des Ingénieurs-Conseils, 2017). Upon project completion ($P_{i,t}^{\text{actual}} = 1$), the accumulated retention $R_i^{\text{ret}} = \rho_i \cdot CP_i$ is released together with the final milestone payment. The payment identity $A_i + \sum_j R_{i,j}^{\text{net}} + R_i^{\text{ret}} = CP_i$ holds exactly.

Milestone payments are subject to stochastic timing delays between certification and receipt at Levels 2–3 (Aibinu and Jagboro, 2002; Odeh and Battaineh, 2002); the amount is fixed at the contractual net value but the period of receipt is a random variable. Aggregate period inflow is:

$$V_t = \sum_{i,j} R_{i,j}^{\text{net}} \cdot \mathbf{1}\{(i,j) \text{ newly certified at } t\} + \sum_{i \in \mathcal{P}_t^{\text{term}}} R_i^{\text{term}}, \quad (4)$$

where $\mathcal{P}_t^{\text{term}}$ collects projects newly terminated at t , with settlement R_i^{term} defined in Section 3.2.5 below.

3.2.4 Performance Monitoring: EVM Signals

The Earned Value Management framework (Fleming and Koppelman, 2010) provides the diagnostic vocabulary used to monitor contractor performance and to drive the termination mechanic. The Earned Value at period t is $EV_{i,t} = P_{i,t}^{\text{actual}} \cdot BAC_i$ (equivalently, the Budgeted Cost of Work Performed, BCWP), expressed in cost units against the BAC_i baseline. Three derived indices are computed each period:

$$SPI_{i,t} = \frac{P_{i,t}^{\text{actual}}}{P_{i,t}^{\text{plan}}}, \quad (5)$$

$$CPI_{i,t} = \frac{EV_{i,t}}{ACWP_{i,t}} = \frac{P_{i,t}^{\text{actual}} \cdot BAC_i}{ACWP_{i,t}}, \quad (6)$$

$$EAC_{i,t} = ACWP_{i,t} + \frac{BAC_i - EV_{i,t}}{CPI_{i,t}}. \quad (7)$$

$SPI_{i,t} < 1$ signals schedule lag; $CPI_{i,t} < 1$ signals cost overrun. $EAC_{i,t}$ is the forward-looking Estimate at Completion: it projects the total cost of the project at the current cost performance rate by scaling remaining planned work by the inverse of $CPI_{i,t}$, directly analogous to the forecast finish date computation used in Condition 1 of the termination mechanic (Section 3.2.5). ? established that this formulation produces the most accurate and stable cost-at-completion forecasts, with reliability increasing beyond the 20% completion mark (?). All three indices are observable state variables; their stochastic dynamics at Levels 2–3 are calibrated from empirical EVM datasets (Acebes et al., 2014; Batselier and Vanhoucke, 2017).

3.2.5 Termination Mechanic

Termination is an endogenous environment mechanic, not an agent action. At each period t , the environment evaluates two conditions for every active project i . Both are forward-looking trajectory estimates, symmetric in structure to one another.

Condition 1 — Schedule tolerance exhausted. The forecast finish date, computed by linear extrapolation of remaining planned duration at the current schedule performance rate, is:

$$\hat{f}_{i,t} = t + \frac{D_i^{\text{plan}} - (t - s_i)}{\text{SPI}_{i,t}}. \quad (8)$$

The schedule deviation is $\Delta f_{i,t} = \hat{f}_{i,t} - f_i$. Condition 1 activates when the projected delay exceeds the contractual schedule tolerance Ω_i (expressed in periods):

$$\Delta f_{i,t} > \Omega_i. \quad (9)$$

Condition 2 — Cost overrun trajectory unacceptable. The Estimate at Completion $\text{EAC}_{i,t}$ defined in (7) projects the total project cost at the current cost performance rate, by scaling remaining planned work by the inverse of $\text{CPI}_{i,t}$ — directly analogous to the schedule extrapolation in (8). Condition 2 activates when the projected total cost exceeds the project-specific maximum tolerable cost overrun:

$$\text{EAC}_{i,t} > (1 + \mu_i) \cdot \text{BAC}_i, \quad (10)$$

where $\mu_i \geq 0$ is the contractually set maximum cost overrun ratio for project i , fixed at commitment.

Cure-period counter and termination execution. When both conditions hold simultaneously, the environment decrements the project-specific cure-period counter, modelling the FIDIC Sub-Clause 15.1 obligation to allow a recovery interval before termination may be executed ([Fédération Internationale des Ingénieurs-Conseils, 2017](#); [Ndekugri and Rycroft, 2008](#)):

$$\tau_{i,t}^{\text{rem}} = \begin{cases} \tau_{i,t-1}^{\text{rem}} - 1 & \text{if (9) and (10) both hold,} \\ \tau_i^{\text{tol}} & \text{if either condition lapses.} \end{cases} \quad (11)$$

The counter initialises at $\tau_{i,s_i}^{\text{rem}} = \tau_i^{\text{tol}}$, where $\tau_i^{\text{tol}} \in \mathbb{Z}^+$ is the project-specific tolerance, fixed at commitment. Termination executes at the period immediately following $t_i^{\text{term}} = \min\{t : \tau_{i,t}^{\text{rem}} = 0\}$. The counter is part of the observable portfolio state; the agent cannot override its dynamics but influences both conditions indirectly through allocation decisions that drive $\text{SPI}_{i,t}$ and $\text{CPI}_{i,t}$.

Upon termination at t_i^{term} , the net settlement payment is:

$$R_i^{\text{term}} = P_{i,t_i^{\text{term}}}^{\text{actual}} \cdot \text{CP}_i(1 - \rho_i) - \left(A_i - \sum_{j \in \mathcal{M}_i^{\text{cert}}(t_i^{\text{term}})} \delta_i^{\text{rec}} \cdot \phi_{i,j} \cdot \text{CP}_i \cdot \mathbf{1}\left\{ \sum_{k \leq j} \phi_{i,k} \geq \psi_i \right\} \right). \quad (12)$$

The first term is the value of work delivered net of retention; the second is the outstanding advance not yet recovered through milestone deductions. Negative R_i^{term} constitutes a net contractor obligation, recovered from portfolio inflows in subsequent periods consistent with FIDIC termination clauses ([Fédération Internationale des Ingénieurs-Conseils, 2017](#)). The settlement formula values delivered work at $P_{i,t_i^{\text{term}}}^{\text{actual}} \cdot \text{CP}_i(1 - \rho_i)$ — the fraction of contract price attributable to physically completed scope, net of retention. This is not a penalty construct; it is a direct valuation of productive assets transferred at termination ([Dixit and Pindyck, 1994](#)), grounding the model's financial signals in the real asset value of work in place rather than in arbitrarily weighted deviation terms. A portfolio manager interpreting a negative R_i^{term} correctly reads it as the cost of having disbursed an advance against work that was never delivered — a capital allocation error whose financial consequence is fully determined by the contract without requiring external calibration.

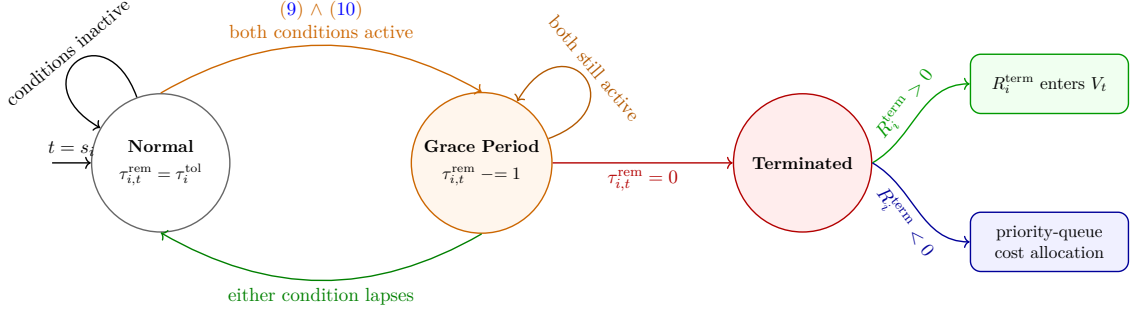


Figure 5: Finite-state representation of the termination mechanic for project i .

3.2.6 Portfolio Cash Balance

The portfolio cash balance evolves as:

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t, \quad B_t \geq 0 \quad \forall t \in \mathcal{T}. \quad (13)$$

The constraint $B_t \geq 0$ prohibits borrowing; the portfolio operates under a fixed capital commitment (Kavadias and Loch, 2003). The shared balance B_t is the sole coupling across otherwise independent project dynamics.

3.2.7 Observable State and Shared Objective

The portfolio state at period t is:

$$S_t = \left(\{P_{i,t}^{\text{actual}}, \text{ACWP}_{i,t}, \text{SPI}_{i,t}, \text{CPI}_{i,t}, \text{EAC}_{i,t}, \Delta f_{i,t}, \text{status}_{i,t}, \tau_{i,t}^{\text{rem}}\}_{i \in \mathcal{P}}, B_t \right), \quad (14)$$

where $\text{status}_{i,t} \in \{\text{active}, \text{completed}, \text{terminated}\}$. This gives a state dimension of $8n + 1$ per period. All components are observable; $\eta_{i,t}$ is the sole unobservable primitive, realised only after $x_{i,t}$ is committed.

The decision-maker maximises discounted net cash flow over the horizon:

$$\max_{\{x_{i,t}\}} \mathbb{E} \left[\sum_{t=1}^H \gamma^{t-1} \left(V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \right) \right], \quad (15)$$

subject to, for all $i \in \mathcal{P}$, $t \in \mathcal{T}$:

$$x_{i,t} \geq 0, \quad (16)$$

$$\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t, \quad (17)$$

$$x_{i,t} = 0 \quad \text{if } \text{status}_{i,t} \neq \text{active}. \quad (18)$$

There is no terminal sum and no deviation penalty in the objective. Completion incentives are embedded structurally: R_i^{ret} and R_{i,M_i}^{net} enter V_t at the final certification period; R_i^{term} (possibly negative) enters at termination, creating a natural deterrent against allowing project performance to deteriorate to the termination threshold (Ng et al., 1999). Symmetrically, the discounting of inflows by γ^{t-1} in (15) creates a completion incentive from above — earlier certification of milestone j yields $R_{i,j}^{\text{net}}$ at a higher discount weight, so

a return-maximising agent will front-load allocations to projects where earlier certification is feasible, without any additional objective term. In competitive project-based markets where π_i is fixed at contract signing, this discounting effect is the formal expression of the capital velocity argument: the annualised return on committed capital increases with completion speed, and γ encodes the opportunity cost of capital that makes this concrete (?). The expectation $\mathbb{E}[\cdot]$ collapses to a deterministic evaluation at Level 1 and is retained at Levels 2 and 3.

3.3 Consolidated Parameter Reference

Table 2 lists all project parameters fixed at commitment, grouped by function. The planned S-curve and advance recovery parameters introduced in Sections 3.2.2–3.2.3 are included for completeness. The discount factor γ and initial balance B_1 are portfolio-level parameters shared across all projects.

3.4 Level 1: Deterministic Full-Foresight MILP

Setting $\eta_{i,t} = \bar{\eta}_{i,t}$ (known) renders S_t a deterministic function of $\mathbf{x}_t$, so $\mathcal{P}_t^{\text{active}}$, milestone certification times, and termination periods t_i^{term} are all computable ex ante. Under deterministic efficiency, the trajectories of $\text{SPI}_{i,t}$, $\text{CPI}_{i,t}$, $\text{EAC}_{i,t}$, and the counter $\tau_{i,t}^{\text{rem}}$ are fully determined before solving; both termination conditions (9)–(10) and the counter dynamics (11) are parameter-verifiable, so termination periods t_i^{term} and their settlements R_i^{term} enter the MILP as fixed parameters rather than optimisation constraints.

Introducing the certification binary $u_{i,j,t} \in \{0,1\}$ (= 1 iff milestone j of project i certified by period t) and activity indicator $\text{act}_{i,t} \in \{0,1\}$ (fixed parameter), the stochastic objective (15) collapses to:

$$\max \quad \sum_{t=1}^H \gamma^{t-1} \left(V_t^{\text{det}} - \sum_i \text{act}_{i,t} x_{i,t} \right) \quad (19a)$$

$$\text{s.t.} \quad B_{t+1} = B_t - \sum_i \text{act}_{i,t} x_{i,t} + V_t^{\text{det}}, \quad B_t \geq 0, \quad \forall t \quad (19b)$$

$$\sum_i \text{act}_{i,t} x_{i,t} \leq B_t, \quad \forall t \quad (19c)$$

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \bar{\eta}_{i,t} \frac{x_{i,t}}{\text{BAC}_i}, \quad P_{i,t}^{\text{actual}} \leq 1, \quad \forall i, t \quad (19d)$$

$$P_{i,t}^{\text{actual}} \geq \theta_{i,j} u_{i,j,t} - \mathbf{M}(1 - u_{i,j,t}), \quad \forall i, j, t \quad (19e)$$

$$P_{i,t}^{\text{actual}} \leq \theta_{i,j} + \mathbf{M} u_{i,j,t}, \quad u_{i,j,t} \geq u_{i,j,t-1}, \quad u_{i,j+1,t} \leq u_{i,j,t}, \quad \forall i, j, t \quad (19f)$$

$$u_{i,j,t} = 0 \quad \text{if } t < e_{i,j}, \quad \forall i, j, t \quad (19g)$$

$$\begin{aligned} V_t^{\text{det}} = & \sum_{i,j} R_{i,j}^{\text{net}} (u_{i,j,t} - u_{i,j,t-1}) + \sum_i R_i^{\text{ret}} (u_{i,M_i,t} - u_{i,M_i,t-1}) \\ & + \sum_i A_i \cdot \mathbf{1}\{t = s_i\} + \sum_{i \in \mathcal{P}^{\text{term}}} R_i^{\text{term}} \cdot \mathbf{1}\{t = t_i^{\text{term}}\}, \quad \forall t \end{aligned} \quad (19h)$$

$$x_{i,t} = 0 \quad \text{if } \text{act}_{i,t} = 0, \quad \forall i, t \quad (19i)$$

$$x_{i,t} \geq 0, \quad u_{i,j,t} \in \{0,1\}, \quad u_{i,j,0} = P_{i,0}^{\text{actual}} = 0. \quad (19j)$$

The big- M constant is $\mathbf{M} = \max_{i,j} \theta_{i,j}$. Constraints (19e)–(19f) implement exact if-and-only-if certification logic with monotone, ordered binary activation. Constraint (19g)

Parameter	Symbol	Domain	Definition
<i>Schedule</i>			
Planned start & finish	s_i, f_i	$\mathbb{Z}^+$	Anchor of execution window and S-curve.
Planned duration	D_i^{plan}	$\mathbb{Z}^+$	$f_i - s_i + 1$; normalises progress against plan.
S-curve shape	a_i, b_i	$\mathbb{R}^+$	Beta CDF parameters governing planned expenditure profile (2).
Milestone threshold	$\theta_{i,j}$	$(0, 1]$	Cumulative progress fraction required to certify milestone j .
Earliest certification	$e_{i,j}$	$\mathbb{Z}^+$	Earliest period at which milestone j may be certified, regardless of progress achieved; enforces FIDIC Sub-Clause 14.3 payment interval constraints (Fédération Internationale des Ingénieurs-Conseils, 2017).
Schedule tolerance	Ω_i	$\mathbb{Z}^+$	Maximum admissible forecast finish overrun in periods before Condition 1 (9) activates.
<i>Contract</i>			
Contract price	CP_i	$\mathbb{R}^+$	Total client payment; $CP_i = \text{BAC}_i(1 + \pi_i)$; payment accumulation baseline only.
Budget at completion	BAC_i	$\mathbb{R}^+$	Total planned contractor expenditure; EVM cost basis and progress normalisation denominator.
Profit margin	π_i	$\mathbb{R}_0^+$	Contractual markup on BAC_i .
Advance payment ratio	α_i	$[0, 1]$	$A_i = \alpha_i \cdot CP_i$ received at $t = s_i$.
Recovery deduction rate	δ_i^{rec}	$(0, 1]$	Fraction of each interim payment deducted toward advance recovery once threshold ψ_i is exceeded; default 0.25 per FIDIC Sub-Clause 14.2 (Fédération Internationale des Ingénieurs-Conseils, 2017).
Recovery threshold	ψ_i	$[0, 1]$	Cumulative certified payments as fraction of CP_i at which advance recovery begins; default 0.10 per FIDIC Sub-Clause 14.2 (Fédération Internationale des Ingénieurs-Conseils, 2017).
Retention ratio	ρ_i	$[0, 1]$	Fraction withheld per milestone; released as $\rho_i \cdot CP_i$ at completion.
Milestone weight	$\phi_{i,j}$	$\mathbb{R}^+$	Fraction of CP_i at milestone j ; $\sum_j \phi_{i,j} = 1$.
<i>Termination</i>			
Max cost overrun ratio	μ_i	$\mathbb{R}_0^+$	Contractual ceiling on projected total cost expressed as fractional overrun on BAC_i ; activates Condition 2 (10).
Termination tolerance	τ_i^{tol}	$\mathbb{Z}^+$	Cure-period length in periods; models FIDIC Sub-Clause 15.1 (Fédération Internationale des Ingénieurs-Conseils, 2017; Ndekugri and Rycroft, 2008).
<i>Portfolio</i>			
Discount factor	γ	$(0, 1]$	Per-period discount factor; encodes time preference and opportunity cost of capital (?); shared across $\mathcal{P}$.
Initial cash balance	B_1	$\mathbb{R}^+$	Portfolio cash balance at horizon start.
Planning horizon	H	$\mathbb{Z}^+$	Derived as $\max_i f_i - \min_i s_i + 1$; $t = 1$ aligns with start of first period.

enforces the contractual earliest certification date: $u_{i,j,t}$ is fixed to zero for all periods prior to $e_{i,j}$, preventing the MILP from certifying a milestone before its contractual window opens regardless of how much budget has been allocated. This makes the optimal allocation timing uniquely determined under $\gamma < 1$: the agent will front-load spend to reach $\theta_{i,j}$ exactly at $e_{i,j}$, then hold until the window opens. The difference $(u_{i,j,t} - u_{i,j,t-1})$ in (19h) equals one in the sole period a milestone is first certified, linearising the payment trigger without additional binaries.

Upper bound. Let Z_{L1}^* denote the Level 1 optimum. For any causal policy π feasible under uncertainty, $\sup_{\pi} \mathbb{E}[Z^{\pi}] \leq Z_{L1}^*$: every feasible causal allocation sequence is also feasible in the Level 1 program solved on the realised trajectory, so the full-foresight optimum dominates in expectation. The gap $Z_{L1}^* - Z^{\pi}$ quantifies the cost of uncertainty and anchors the empirical comparison in Section ??.

3.5 Level 2: Stochastic Extension and Intractability

3.5.1 Stochastic Primitive and Distributional Assumption

At Level 2, the deterministic efficiency scalar $\bar{\eta}_{i,t}$ is replaced by a random variable $\eta_{i,t}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The efficiency realisations are assumed to satisfy:

Assumption 1 (Efficiency structure). For each project $i \in \mathcal{P}$ and period $t \in \mathcal{T}$:

- (i) $\eta_{i,t} > 0$ almost surely;
- (ii) $\{\eta_{i,t}\}_{t \in \mathcal{T}}$ are independent across periods given project i ;
- (iii) $\eta_{i,t}$ are independent across projects $i \in \mathcal{P}$;
- (iv) the marginal distribution of $\eta_{i,t}$ is project-specific and calibrated from empirical EVM datasets (Acebes et al., 2014; Batselier and Vanhoucke, 2017).

The realisation $\eta_{i,t}(\omega)$ for scenario $\omega \in \Omega$ is revealed to neither the portfolio manager nor the environment until *after* the allocation $x_{i,t}$ is committed. This timing assumption is the defining structural difference from Level 1: the manager must act under genuine uncertainty about the period's productivity before observing its outcome.

3.5.2 Information Structure and Non-Anticipativity

Let $\mathcal{F}_t$ denote the σ -algebra generated by all efficiency realisations and resulting state transitions up to and including period t :

$$\mathcal{F}_t = \sigma(\eta_{i,\tau}(\omega) : i \in \mathcal{P}, \tau \leq t), \quad \mathcal{F}_0 = \{\emptyset, \Omega\}. \quad (20)$$

The history available to the portfolio manager at the start of period t , prior to committing $\mathbf{x}_t$, is:

$$\mathcal{H}_t = (S_1, \mathbf{x}_1, S_2, \mathbf{x}_2, \dots, S_{t-1}, \mathbf{x}_{t-1}, S_t), \quad (21)$$

where S_t is the portfolio state defined in (14), which is $\mathcal{F}_{t-1}$ -measurable since it reflects only realisations up to period $t-1$. The *non-anticipativity constraint* requires that the allocation at period t depends only on the available history:

$$\mathbf{x}_t \in \sigma(\mathcal{H}_t) \quad \forall t \in \mathcal{T}. \quad (22)$$

This constraint distinguishes a causal policy from the full-foresight MILP of Level 1, which optimises over the entire realised trajectory $(\eta_{i,t})_{i,t}$ simultaneously.

3.5.3 The Multi-Stage Stochastic Program

Under Assumption 1 and constraint (22), the Level 1 MILP (19) becomes a multi-stage stochastic program (Birge and Louveaux, 2011). The decision at each stage is a *policy* $\pi_t : \mathcal{H}_t \rightarrow \mathbb{R}_{\geq 0}^{|\mathcal{P}_t^{\text{active}}|}$ rather than a fixed scalar, and the objective is an expectation over all scenario realisations. The program is:

$$\max_{\{\pi_t\}} \mathbb{E} \left[\sum_{t=1}^H \gamma^{t-1} \left(V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \right) \right] \quad (23a)$$

$$\text{s.t. } x_{i,t} = \pi_t(\mathcal{H}_t)_i, \quad \mathbf{x}_t \in \sigma(\mathcal{H}_t), \quad \forall i, t \quad (23b)$$

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \eta_{i,t} \cdot \frac{x_{i,t}}{\text{BAC}_i}, \quad \forall i, t \quad (23c)$$

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t, \quad B_t \geq 0, \quad \forall t \quad (23d)$$

$$\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t, \quad \forall t \quad (23e)$$

$$x_{i,t} = 0 \quad \text{if } \text{status}_{i,t} \neq \text{active}, \quad \forall i, t \quad (23f)$$

$$x_{i,t} \geq 0, \quad \forall i, t \quad (23g)$$

where V_t is as defined in (4), now a random variable whose realisation depends on $(\eta_{i,\tau})_{\tau \leq t}$ through the progress recursion (23c) and the milestone certification events it generates. The termination counter (11), and therefore $\mathcal{P}_t^{\text{active}}$ and $\text{status}_{i,t}$, are likewise random variables endogenous to the realised efficiency trajectory, in contrast to Level 1 where they were fixed parameters.

3.5.4 Scenario-Tree Representation and Exponential Growth

The canonical solution method for multi-stage stochastic programs is scenario-tree enumeration (Birge and Louveaux, 2011). The scenario tree for (23) has the following structure. Discretise $\eta_{i,t}$ to K values per project per period. At each period t , the tree branches by K^n (one branching factor of K per project, with n projects independent by Assumption 1(iii)). Over a horizon of H periods the tree has:

$$|\mathcal{S}| = K^{nH} \quad (24)$$

leaf nodes, each corresponding to a complete realisation of $(\eta_{i,t})_{i \in \mathcal{P}, t \in \mathcal{T}}$.

For a representative infrastructure portfolio with $n = 10$ projects, a horizon H derived endogenously from the committed portfolio (typically $H \in [18, 36]$ months in practice), and $K = 5$ discrete efficiency scenarios per project per period, the leaf-node count is:

$$|\mathcal{S}| = 5^{10 \times 24} = 5^{240} \approx 10^{167}, \quad (25)$$

a number that exceeds the estimated number of atoms in the observable universe by more than 80 orders of magnitude. This is not a consequence of the binary milestone variables inherited from the MILP structure: replacing them with continuous relaxations leaves the exponent nH unchanged, since the exponential growth is intrinsic to the multi-stage stochastic formulation and independent of the within-stage problem structure.

Endogenous horizon compounding. A further compounding factor distinguishes this problem from standard multi-stage stochastic programs with fixed horizons. The planning horizon $H = \max_i f_i - \min_i s_i + 1$ is itself a function of the committed portfolio; it is not an external configuration parameter. Under stochastic efficiency, the effective execution window of each project is uncertain — a project with $\text{SPI}_{i,t} < 1$ throughout will overrun f_i , extending the realised horizon beyond H . In the worst case, every project overruns simultaneously, and the actual number of periods over which decisions must be made grows with the portfolio. Each additional period multiplies the scenario tree by K^n , so horizon uncertainty compounds (24) multiplicatively.

Termination as structural bifurcation. At Level 1, termination periods t_i^{term} were fixed parameters precomputed from the known efficiency trajectory. At Level 2, the period at which the cure-period counter (11) reaches zero depends on the realised sequence of $\text{SPI}_{i,t}$ and $\text{CPI}_{i,t}$ values, which are in turn determined by $(\eta_{i,\tau})_{\tau \leq t}$. Each termination event removes project i from $\mathcal{P}_t^{\text{active}}$ and delivers R_i^{term} into V_t — a structural change in the active project set that bifurcates the scenario tree at every period in which a project’s counter is positive. Across n projects each with counter depth τ_i^{tol} , the number of structurally distinct subtrees grows combinatorially in the termination event space, adding a second source of exponential scaling beyond (24).

3.5.5 Intractability and the Motivation for Level 3

The exponential scaling established in (24) constitutes a fundamental computational barrier. It is not a function of solver technology or implementation efficiency; it is intrinsic to the multi-stage stochastic formulation of any sequential decision problem with independent per-stage uncertainty across n components over H stages (Birge and Louveaux, 2011). Approximation methods for multi-stage stochastic programs — Sample Average Approximation (Kleywegt et al., 2002), Progressive Hedging (Rockafellar and Wets, 1991), and Stochastic Dual Dynamic Programming (Pereira and Pinto, 1991) — reduce the computational burden but do not escape the exponential scaling in n and H at portfolio scales of practical interest. Sample Average Approximation, for instance, requires a sample size that grows with the variance of the objective and the number of stages to achieve a given optimality gap (Kleywegt et al., 2002); for $n = 10$ and $H = 24$ this remains intractable without structural decomposition that the shared cash balance (23d) prevents.

Proposition 1 (Level 1 as upper bound). *Let Z_{L1}^* denote the optimal value of the Level 1 MILP (19) evaluated on a realised efficiency trajectory $(\bar{\eta}_{i,t})_{i,t}$. Let Z^π denote the objective value achieved by any causal policy π satisfying (22) on the same trajectory. Then:*

$$\sup_{\pi \in \Pi^{\text{causal}}} \mathbb{E}[Z^\pi] \leq Z_{L1}^*, \quad (26)$$

where Π^{causal} denotes the set of all policies satisfying (22).

Proof. Fix any realised trajectory $\omega \in \Omega$. Any allocation sequence $\{x_{i,t}\}$ produced by a causal policy π under ω is a feasible solution to the Level 1 MILP solved on ω (since feasibility constraints (19b)–(19i) are identical to (23d)–(23f) for fixed $\bar{\eta}_{i,t} = \eta_{i,t}(\omega)$). Therefore $Z^\pi(\omega) \leq Z_{L1}^*(\omega)$ for every ω . Taking expectations on both sides yields (26). $\square$

The gap $Z_{L1}^* - \mathbb{E}[Z^\pi]$ quantifies the cost of uncertainty: the value destroyed by operating under incomplete information relative to a hypothetical full-foresight oracle. This gap is the primary empirical quantity reported in Section ?? . The intractability of computing

the exact solution to (23) motivates the Level 3 reformulation: rather than enumerating the scenario tree, the problem is recast as a sequential interaction between a policy and an environment, solved through sampled trajectory experience rather than exhaustive enumeration.

3.6 Level 3: Constrained MDP and RL Agent

The intractability established in Section 3.5 rules out exact solution of (23) at realistic portfolio scales. The Level 3 reformulation replaces exhaustive scenario-tree enumeration with a sequential interaction framework: the portfolio manager is modelled as a *policy* that observes the current state, commits an allocation, and receives a reward signal, with the environment transitioning stochastically to the next state. Optimisation proceeds through repeated sampled interaction rather than explicit enumeration, making the computational cost per training step independent of $|\mathcal{S}| = K^{nH}$.

3.6.1 Constrained Markov Decision Process Formulation

Definition 1 (Portfolio Governance CMDP). The portfolio budget allocation problem at Level 3 is formalised as a finite-horizon Constrained Markov Decision Process (Altman, 1999):

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, R, \mathcal{C}, \gamma, H), \quad (27)$$

with components defined as follows.

State space $\mathcal{S}$. The state at period t is the portfolio state S_t defined in (14):

$$S_t \in \mathcal{S} \subseteq [0, 1]^n \times \mathbb{R}_{\geq 0}^n \times \mathbb{R}^{4n} \times \{0, 1, 2\}^n \times \mathbb{Z}_{\geq 0}^n \times \mathbb{R}_{\geq 0}, \quad (28)$$

collecting, for each project i : actual progress $P_{i,t}^{\text{actual}} \in [0, 1]$, cumulative actual cost $\text{ACWP}_{i,t} \in \mathbb{R}_{\geq 0}$, performance indices $\text{SPI}_{i,t}$, $\text{CPI}_{i,t}$, $\text{EAC}_{i,t}$, schedule deviation $\Delta f_{i,t}$, project status $\text{status}_{i,t} \in \{0, 1, 2\}$ encoding `{active, completed, terminated}`, and cure-period counter $\tau_{i,t}^{\text{rem}} \in \mathbb{Z}_{\geq 0}$; together with the portfolio cash balance $B_t \in \mathbb{R}_{\geq 0}$. The state dimension is $8n + 1$. The sole unobservable primitive is $\eta_{i,t}$, realised only after $\mathbf{x}_t$ is committed.

Action space $\mathcal{A}$ and feasible set. The raw action space is:

$$\mathcal{A} = \mathbb{R}_{\geq 0}^n, \quad (29)$$

with the state-dependent feasible action set:

$$\mathcal{A}(S_t) = \left\{ \mathbf{x}_t \in \mathbb{R}_{\geq 0}^n \left| \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t, \quad x_{i,t} = 0 \text{ if } \text{status}_{i,t} \neq \text{active} \right. \right\}, \quad (30)$$

encoding constraints (17) and (18) directly. The hard liquidity constraint $B_t \geq 0$ is enforced by construction through the projection operator defined below.

Projection operator. To guarantee $\mathbf{x}_t \in \mathcal{A}(S_t)$ for any raw network output $\tilde{\mathbf{x}}_t \in \mathbb{R}^n$, the environment applies a two-step projection before executing the transition. First, allocations to inactive projects are zeroed: $\tilde{x}_{i,t} \leftarrow 0$ for all i with $\text{status}_{i,t} \neq \text{active}$. Second, the remaining vector is projected onto the budget simplex:

$$\mathbf{x}_t = B_t \cdot \frac{\max(\tilde{\mathbf{x}}_t^{\text{active}}, \mathbf{0})}{\|\max(\tilde{\mathbf{x}}_t^{\text{active}}, \mathbf{0})\|_1 \vee 1}, \quad (31)$$

where $\tilde{\mathbf{x}}_t^{\text{active}}$ denotes the sub-vector restricted to active projects, and $\vee$ denotes the maximum operator ensuring the denominator is at least one when the agent outputs all-zero or all-negative values. This projection is differentiable almost everywhere and preserves gradient flow through the policy network during training (Schulman et al., 2017).

Transition kernel $\mathcal{T}$. The stochastic transition $\mathcal{T}(S_{t+1} | S_t, \mathbf{x}_t)$ is defined by the composition of environment mechanics applied sequentially within each period:

- (i) Draw $\eta_{i,t} \sim F_i$ independently for each active project i , where F_i is the project-specific efficiency distribution calibrated from Acebes et al. (2014); Batselier and Vanhoucke (2017).
- (ii) Update progress and actual cost via (1).
- (iii) Compute EVM signals via (5)–(7).
- (iv) Evaluate termination conditions (9)–(10) and update counter via (11).
- (v) Certify milestones for which $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$ and $t \geq e_{i,j}$; compute V_t via (3)–(4); apply stochastic payment delays (Aibinu and Jagboro, 2002).
- (vi) Update portfolio cash balance via (13).
- (vii) Execute termination for any project with $\tau_{i,t}^{\text{rem}} = 0$, compute R_i^{term} via (12).

The transition kernel is Markovian: S_{t+1} depends on $(S_t, \mathbf{x}_t, \boldsymbol{\eta}_t)$ only, not on $\mathcal{H}_{t-1}$. This is the key structural property that makes the CMDP formulation exact rather than approximate: the full portfolio state S_t defined in (14) is a sufficient statistic for the history $\mathcal{H}_t$ defined in (21).

Reward signal R . The per-period reward is the discounted net cash flow term from (15):

$$r_t(S_t, \mathbf{x}_t, S_{t+1}) = \gamma^{t-1} \left(V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \right), \quad (32)$$

where V_t is realised after the transition. The reward contains no deviation penalty term; completion incentives are embedded structurally through R_i^{ret} and R_{i,M_i}^{net} at project completion and through R_i^{term} (possibly negative) at termination, as established in Section 3.2.7.

Constraint set $\mathcal{C}$. The hard liquidity constraint $B_t \geq 0$ is the sole constraint promoted to the CMDP constraint set:

$$\mathcal{C} = \{\mathbf{x}_t \in \mathcal{A} \mid B_{t+1} \geq 0\}. \quad (33)$$

This constraint is enforced by the projection (31) rather than through a Lagrangian penalty, ensuring hard satisfaction at every step rather than satisfaction only in expectation (Altman, 1999).

CMDP objective. The policy optimisation problem is:

$$\pi^* = \operatorname{argmax}_{\pi \in \Pi} J(\pi) = \operatorname{argmax}_{\pi \in \Pi} \mathbb{E}_{\pi} \left[\sum_{t=1}^H r_t(S_t, \mathbf{x}_t, S_{t+1}) \right], \quad (34)$$

subject to $\mathbf{x}_t \in \mathcal{C}$ for all t , where Π denotes the set of all measurable policies satisfying (22) and $\mathbb{E}_{\pi}$ denotes expectation under the distribution over trajectories induced by π and $\mathcal{T}$. The objective $J(\pi)$ in (34) is identical to the intractable stochastic program (23a); the CMDP reformulation does not change the objective but changes the solution method from scenario-tree enumeration to policy-gradient optimisation.

Remark 1 (Relationship to Level 1 upper bound). By Proposition 1, $J(\pi^*) \leq Z_{L1}^*$ for any causal policy π^* . The CMDP does not close this gap; it provides the tractable path to finding the best achievable $J(\pi)$ within Π^{causal} . The empirical gap $Z_{L1}^* - J(\hat{\pi})$, where $\hat{\pi}$ is the trained PPO agent, quantifies the combined cost of uncertainty and approximation error in Section ??.

3.6.2 Gymnasium-Compatible Environment

The CMDP $\mathcal{M}$ is implemented as a configurable Gymnasium-compatible environment (Towers et al., 2024). The environment exposes the standard `reset()` / `step()` interface, with `observation_space` = $\mathcal{S}$ and `action_space` = $\mathcal{A}$. All project and portfolio parameters (Table 2) are passed as a configuration dictionary at initialisation; the planning horizon H is derived automatically as $\max_i f_i - \min_i s_i + 1$ from the committed project set, making the environment fully parameterisable across contract types and portfolio sizes without code modification.

The observation vector returned at each step is a flattened concatenation of S_t as defined in (28), normalised to $[0, 1]$ per component to facilitate stable policy gradient estimation. The environment enforces the projection (31) on the raw action vector before applying the transition kernel, guaranteeing $\mathbf{x}_t \in \mathcal{A}(S_t)$ regardless of the policy output. Episodes terminate when $t = H$ or when all projects have reached `completed` or `terminated` status, whichever occurs first.

The stochastic efficiency $\eta_{i,t}$ is drawn each period from a project-specific Beta distribution calibrated from empirical EVM datasets (Acebes et al., 2014; Batselier and Vanhoucke, 2017); the Beta family accommodates the right-skewed, bounded support structure of empirical CPI and SPI distributions documented in ? and Batselier and Vanhoucke (2017). Milestone payment timing delays are drawn from project-specific discrete distributions over $\{0, 1, \dots, d_i^{\max}\}$ periods (Aibinu and Jagboro, 2002), applied to the net payment amount $R_{i,j}^{\text{net}}$ at certification without affecting the certified amount itself.

3.6.3 PPO Agent and Training Protocol

Algorithm selection. Proximal Policy Optimisation (Schulman et al., 2017) is selected as the solution algorithm for three reasons. First, its clipped surrogate objective provides stable updates in continuous action spaces without the step-size sensitivity of vanilla policy gradient methods. Second, its on-policy sample collection is compatible with the non-stationary episode structure of the environment, in which the active project set $\mathcal{P}_t^{\text{active}}$ changes endogenously across periods. Third, PPO has demonstrated consistent performance in operations management domains with similar state-action structure (Giannocaro and Pontrandolfo, 2002; Peng and Huang, 2021).

PPO surrogate objective. At each training iteration, PPO collects a batch of trajectories $\mathcal{D} = \{\tau_k\}_{k=1}^N$ under the current policy π_θ and optimises the clipped surrogate:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_{\mathcal{D}} \left[\min \left(\frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_t, \text{clip} \left(\frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right], \quad (35)$$

where $\hat{A}_t$ is the Generalised Advantage Estimate (Schulman et al., 2016) computed from the value function baseline $V_\phi(S_t)$ trained jointly with the policy, and ϵ is the clipping threshold. The value function shares the lower network layers with the policy (actor-critic architecture), with separate output heads for π_θ and V_ϕ .

Policy architecture. The policy network maps the observation vector $S_t \in \mathbb{R}^{8n+1}$ to a raw allocation vector $\tilde{\mathbf{x}}_t \in \mathbb{R}^n$ through a fully connected network with layer normalisation. The raw output is passed through the projection (31) before being applied to the environment, ensuring constraint satisfaction without a penalty term in (35). The output distribution is a diagonal Gaussian with state-dependent mean $\mu_\theta(S_t)$ and learned log-standard-deviation $\log \sigma_\theta$, consistent with standard continuous-action PPO implementations.

Domain randomisation. To ensure the learned policy $\hat{\pi}$ generalises across project types, contract structures, and market conditions rather than overfitting to a single parameterisation, all parameters in Table 2 are randomised within calibrated ranges at the start of each training episode (Tobin et al., 2017). Formally, at each episode reset, a parameter vector $\xi \sim \mathcal{U}(\Xi)$ is sampled from a calibrated uniform distribution over the parameter space Ξ , and the environment is initialised with ξ . The trained policy $\hat{\pi}$ implicitly conditions on the observable consequences of ξ (as reflected in S_t) rather than on ξ directly, producing a policy that is robust to parameter uncertainty without requiring an explicit model of the parameter distribution at deployment (Tobin et al., 2017).

Validation protocol. Walk-forward validation (Cerqueira et al., 2020) is used as the empirical evaluation protocol to guard against data leakage given the time-series structure of project cash-flow data. The training set consists of episodes whose parameter draws fall within the calibrated historical range; the test set consists of held-out parameter configurations drawn from the tails of Ξ , representing out-of-distribution contract and market conditions. All performance metrics in Section ?? are reported on the held-out test set.

3.6.4 Evaluation Baselines

The trained agent $\hat{\pi}$ is evaluated against three baselines that span the tractability-optimality spectrum established by the three-level arc.

Baseline 1 Level 1 MILP oracle. The MILP (19) is solved on each test episode’s realised efficiency trajectory $(\eta_{i,t}(\omega))_{i,t}$ using a commercial solver. Its objective value $Z_{\text{L1}}^*(\omega)$ is the per-episode upper bound from Proposition 1. The gap $Z_{\text{L1}}^* - J(\hat{\pi})$ is the primary performance metric.

Baseline 2 Two-stage stochastic program. A two-stage approximation of (23) is solved by fixing first-stage decisions $\mathbf{x}_1$ and optimising over the expected recourse across N_s sampled scenarios for subsequent stages (Birge and Louveaux, 2011). This represents

the best tractable exact method available within the stochastic programming tradition and quantifies the value of the full sequential policy over a rolling two-stage approximation.

Baseline 3 Proportional heuristic. At each period t , the available balance B_t is distributed across active projects proportionally to their remaining planned expenditure $\text{BAC}_i(1 - P_{i,t}^{\text{actual}})$:

$$x_{i,t}^{\text{prop}} = B_t \cdot \frac{\text{BAC}_i(1 - P_{i,t}^{\text{actual}})}{\sum_{j \in \mathcal{P}_t^{\text{active}}} \text{BAC}_j(1 - P_{j,t}^{\text{actual}})}. \quad (36)$$

This rule is parameter-free and represents current practice in portfolio management information systems: budget is distributed in proportion to remaining work without regard to contract profitability, schedule urgency, or termination risk. The gap between the PPO agent and this baseline quantifies the practical value of the learned policy over the incumbent approach.

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A Appendix A : PRISMA 2020 Flow Diagram

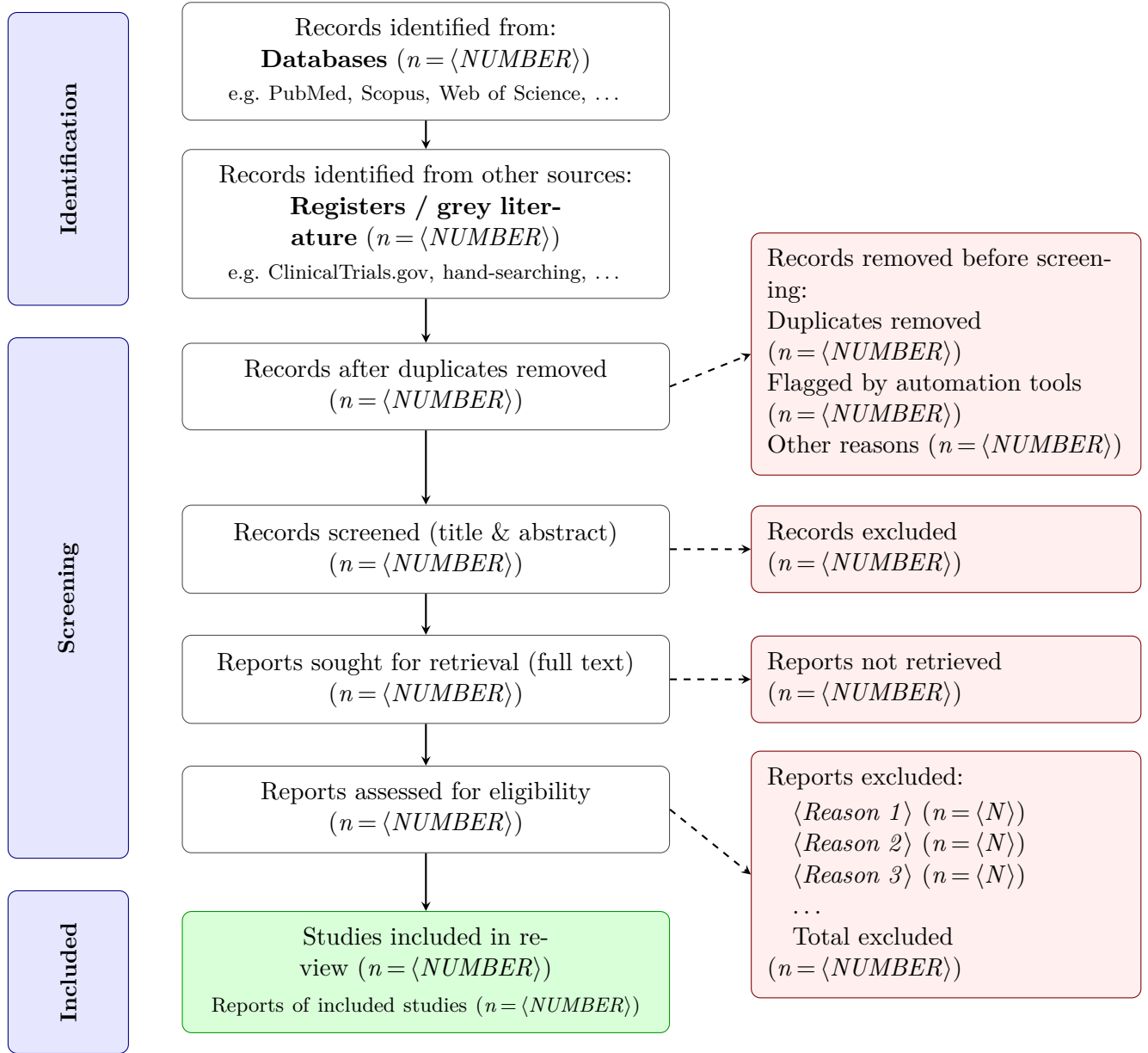


Figure 6: PRISMA 2020 flow diagram of the literature search and selection process.

A Appendix B : Environment verification suite

Conventions

All cases share these conventions unless explicitly overridden:

- Each discrete timestep t represents the interval from the end of period $t - 1$ to the end of period t . All events assigned to period t (spend, progress update, milestone certification, payment) occur at the close of that interval.
- $\gamma = 0.95$ throughout, imposing time value of money in every case.
- Earliest certification dates $e_{i,j}$ are set for every case. The default rule is $e_{i,j} = s_i + \lfloor (j - 1) \cdot D_i^{\text{plan}} / M_i \rfloor$, spreading milestones evenly across the planned duration. Any departure is stated explicitly.
- **Advance payment** is received at $t = s_i - 1$. For single-project cases where $s_i = 1$, this means the advance arrives at $t = 0$ (before the planning horizon begins). The discount factor for the advance term is therefore $\gamma^{s_i - 1 - 1} = \gamma^{-1}$ when the objective is indexed from $t = 1$; equivalently, we index Z from $t = 0$ so the advance contributes $\gamma^0 \cdot A_i$.
- $B_1 = 3 \cdot \sum_i \text{BAC}_i$ (large; cash never binding unless stated).
- $\bar{\eta}_{i,t} = \bar{\eta}$ constant and deterministic unless stated.
- Equal milestone weights $\phi_{i,j} = 1/M_i$ unless stated.
- $\alpha_i = 0$, $\rho_i = 0$ unless stated.
- $\delta^{\text{rec}} = 0.25$, $\psi = 0.10$ (FIDIC defaults) unless stated.
- $\pi_i = 0$ unless stated ($CP_i = \text{BAC}_i$).
- S-curve shape $a_i = b_i = 2$ (symmetric Beta) unless stated.
- Monetary units are abstract.

Note: With $\gamma = 0.95$ and earliest certification dates enforced, the optimal strategy in every completing case is to accumulate progress so that $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$ is first satisfied exactly at $t = e_{i,j}$, then certify immediately when the window opens. Spending earlier than needed wastes discounted cash; spending later than $e_{i,j}$ loses discount value on inflows.

B Group 1: Progress and Milestone Certification

What is being proven: budget converts to progress at rate η , milestones certify at the correct threshold and not before $e_{i,j}$, payments trigger correctly, and milestone ordering is enforced even when progress jumps past multiple thresholds in one period.

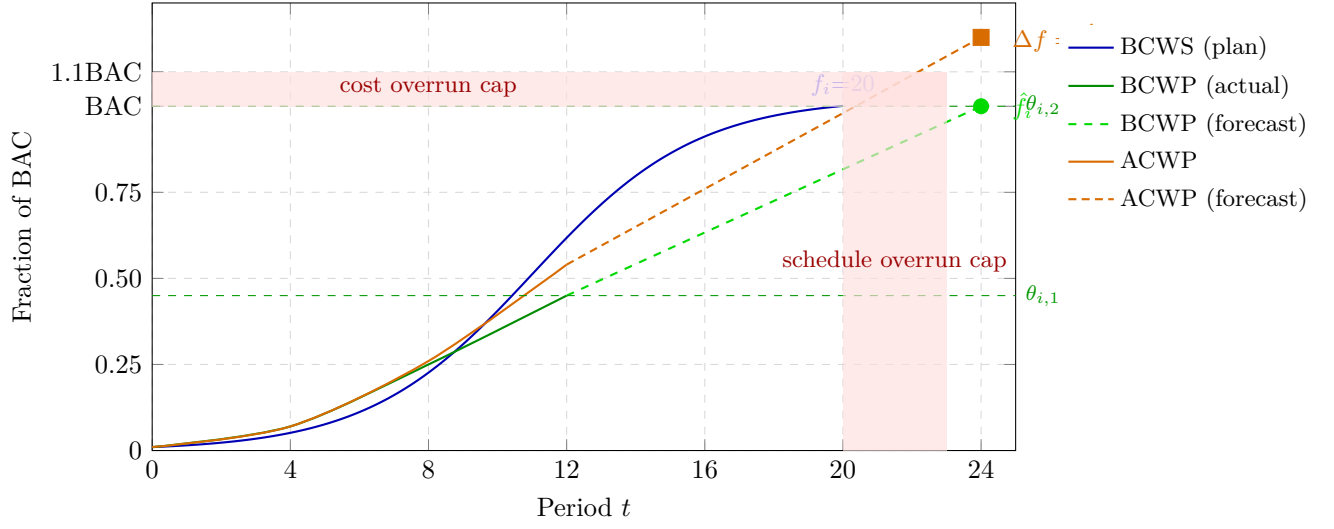


Figure 7: Reference EVM panel style (used as the left panel in every project row). BCWS: planned value; BCWP: earned value; ACWP: actual cost. Green dashed horizontals mark progress thresholds $\theta_{i,j}$. Red shading: schedule and cost overrun caps. $\Delta f = \hat{f}_i - f_i$ is the forecast schedule overrun.

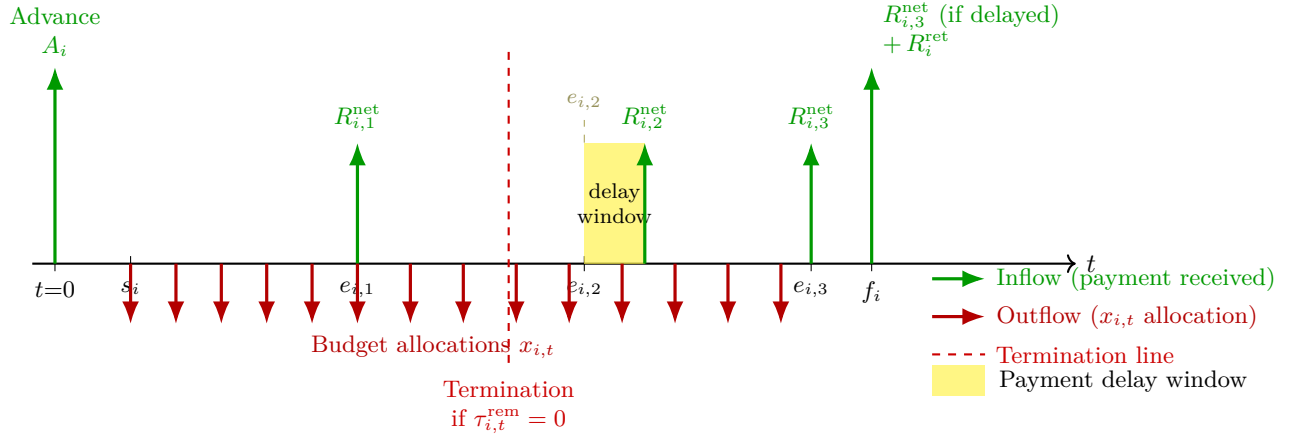


Figure 8: Cash-flow timeline for project i . Advance A_i arrives at $t = s_i - 1$ (period 0 for single-project cases). Green upward arrows: contract inflows. Red downward arrows: budget allocations $x_{i,t}$. Yellow band: stochastic payment delay window between contracted date $e_{i,j}$ and actual receipt. Dashed red vertical: early termination if cure counter exhausted.

B.1 SP-1: Perfect Efficiency, Two Milestones

Economic claim. With $\eta = 1$ and $\gamma = 0.95$, the MILP front-loads spend so that MS1 certifies exactly at $e_{1,1} = 1$. MS2 cannot certify before $e_{1,2} = 3$ regardless of how much progress has been achieved. Deferring the MS2 spend to $t = 3$ saves discounted outflow with identical inflow, so $Z^* = 19.025 > 16.65$ (front-loaded).

Parameters.

$n, s_1, f_1, D^{\text{plan}}$	1, 1, 4, 4
BAC_1, CP_1, π_1	100, 120, 0.20
M_1, θ, ϕ	2, (0.5, 1.0), (0.5, 0.5)
$\bar{\eta}, e_{1,1}, e_{1,2}$	1.0, 1, 3
$\alpha_1, \rho_1, \Omega_1, \mu_1, \tau_1^{\text{tol}}$	0, 0, 1, 0.30, 2
B_1, H, γ	300, 4, 0.95

Milestone payments. $R_{1,1}^{\text{net}} = R_{1,2}^{\text{net}} = 60$.

Optimal allocation.

t	$x_{1,t}$	$P_{1,t}^{\text{actual}}$	Window open?	Inflow
1	50	0.50	MS1: ✓	60
2	0	0.50	MS2: ✗ ($e_{1,2} = 3$)	0
3	50	1.00	MS2: ✓	60
4	0	1.00	—	0

$$Z^* = \gamma^0(60 - 50) + \gamma^1(0) + \gamma^2(60 - 50) = 10 + 0 + 0.9025 \times 10 = \mathbf{19.025}.$$

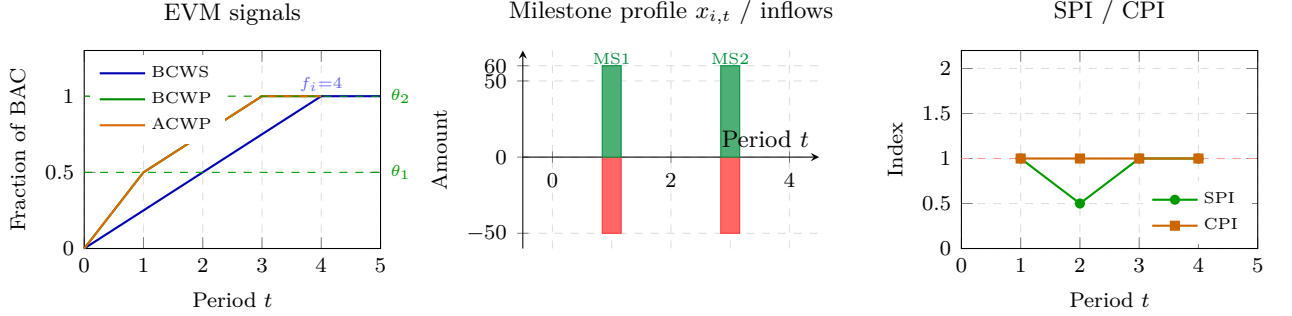


Figure 9: SP-1 project view: EVM signals (left), milestone profile with inflows and allocations (centre), SPI/CPI (right). MS2 allocation correctly deferred to $t = 3$ when the certification window opens.

Assertions for pytest.

- A1. $x_{1,1} = 50 \pm 0.5, x_{1,2} = 0 \pm 0.5, x_{1,3} = 50 \pm 0.5, x_{1,4} = 0$.
- A2. MS1 certified at $t = 1$; MS2 certified at $t = 3$ (not before, since $e_{1,2} = 3$).
- A3. $|Z^* - 19.025| \leq 1.0$.
- A4. Status: `completed`. No τ decrement.

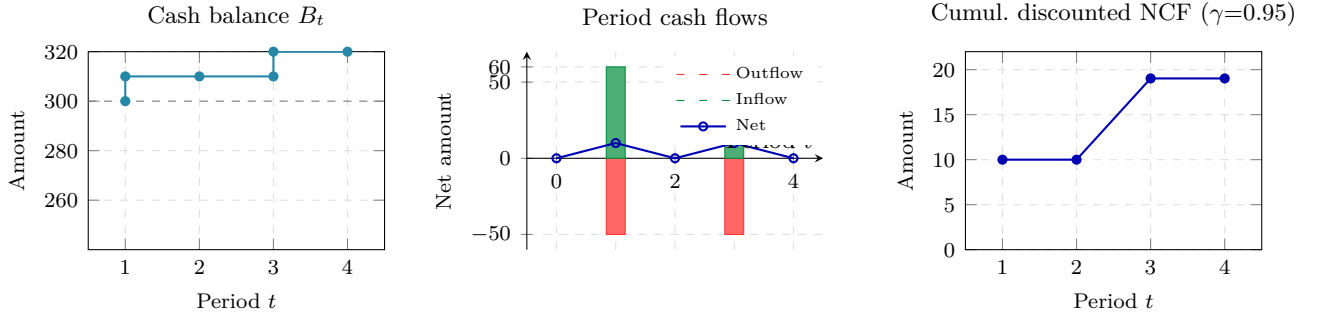


Figure 10: SP-1 portfolio view: cash balance centred at $B_1 = 300$ (left), aggregated period inflows/outflows/net (centre), cumulative discounted NCF ($\gamma=0.95$) (right). $Z^* = 19.025$.

B.2 SP-3: Three Milestones, $\gamma = 0.95$

Economic claim. With $\gamma < 1$ and earliest certification dates, the MILP certifies each milestone exactly when its window opens, spending just enough just in time to meet the progress threshold at each $e_{i,j}$.

Parameters.

$n, s_1, f_1, D^{\text{plan}}$	1, 1, 6, 6
$\text{BAC}_1, \pi_1, CP_1$	60, 0.50, 90
M_1, θ, ϕ	3, (1/3, 2/3, 1), (1/3, 1/3, 1/3)
$\bar{\eta}, e_{1,1}, e_{1,2}, e_{1,3}$	1.0, 1, 3, 5
$\alpha_1, \rho_1, \Omega_1, \mu_1, \tau_1^{\text{tol}}$	0, 0, 2, 0.50, 2

Milestone payments. $R_{1,j}^{\text{net}} = 30$ each.

Optimal allocation. Spend 20 at each $e_{1,j}$.

$$Z^* = \gamma^0 \times 10 + \gamma^2 \times 10 + \gamma^4 \times 10 = 10 + 9.025 + 8.145 = \mathbf{27.17}.$$

Assertions for pytest.

- A1. $x_{1,1} = 20 \pm 0.5, x_{1,3} = 20 \pm 0.5, x_{1,5} = 20 \pm 0.5$; zero otherwise.
- A2. MS j certified at $t \in \{1, 3, 5\}$ respectively.
- A3. $|Z^* - 27.17| \leq 1.0$.
- A4. Status: **completed**.

B.3 SP-J: Milestone Ordering Enforced on Progress Jump

Economic claim. When a single allocation causes progress to clear both thresholds simultaneously, MS2 still cannot certify before $e_{1,2} = 3$ even though $P_{1,1}^{\text{actual}} = 1.0$ qualifies it. The ordering and earliest-date constraints are independent enforcements.

Parameters.

$n, s_1, f_1, D^{\text{plan}}$	1, 1, 4, 4
$\text{BAC}_1, \pi_1, CP_1$	100, 0, 100
M_1, θ, ϕ	2, (0.5, 1.0), (0.5, 0.5)
$\bar{\eta}, e_{1,1}, e_{1,2}$	1.0, 1, 3
$\alpha_1, \rho_1, \Omega_1, \mu_1, \tau_1^{\text{tol}}$	0, 0, 2, 0.50, 2

Optimal strategy. Spend 50 at $t = 1$ (MS1), 50 at $t = 3$ (MS2). All nets zero, so $Z^* = 0$ regardless of discount.

Assertions for pytest.

- A1. MS2 not certified at $t = 1$ or $t = 2$ (window $e_{1,2} = 3$ blocks it).
- A2. MS2 certified at $t = 3$.
- A3. $|Z^* - 0| \leq 1.0$.
- A4. Status: completed by $t = 3$.

C Group 2: Discounting and Timing Incentives

What is being proven: $\gamma < 1$ and earliest certification dates together make the optimal allocation uniquely determined. Deferring outflows to the latest period that still enables certification at $e_{i,j}$ is strictly better than front-loading. When a project can be kept alive by partial spend that freezes the termination counter, survival is optimal if and only if the discounted next milestone inflow exceeds the discounted cost of survival allocations.

C.1 SP-4: Deferring Spend to Last Possible Period is Optimal

Economic claim. Deferring the MS2 spend from $t = 2$ to $t = 5$ (when the window opens) saves $\gamma^1 \times 50 - \gamma^4 \times 50 = 6.775$ in discounted outflow with identical inflow timing.

Parameters.

$n, s_1, f_1, D^{\text{plan}}$	1, 1, 8, 8
$\text{BAC}_1, \pi_1, CP_1$	100, 0.40, 140
M_1, θ, ϕ	2, (0.5, 1), (0.5, 0.5)
$\bar{\eta}, e_{1,1}, e_{1,2}$	1.0, 1, 5
$R_{1,j}^{\text{net}}$	70 each

$$Z^{\text{opt}} = \gamma^0(70 - 50) + \gamma^4(70 - 50) = 20 + 0.8145 \times 20 = \mathbf{36.29}.$$

$$Z^{\text{front}} = \gamma^0(70 - 50) + \gamma^1(-50) + \gamma^4(70) = 29.515 \quad (\text{strictly dominated}).$$

Assertions for pytest.

- A1. $x_{1,1} = 50 \pm 0.5$.
- A2. $x_{1,2} = x_{1,3} = x_{1,4} = 0 \pm 0.5$.
- A3. $x_{1,5} = 50 \pm 0.5$.
- A4. MS1 certified at $t = 1$; MS2 certified at $t = 5$.
- A5. $Z^* \geq 35.5$.
- A6. $Z^* > 29.5$ (front-loaded strategy strictly dominated).

C.2 MP-K: Two Projects, Front-Heavy vs. Back-Heavy Milestones

Economic claim. Project A has a large MS1 payment ($\phi_A = (0.70, 0.30)$) and opens its MS1 window at $e_{A,1} = 1$. Project B has a small MS1 ($\phi_B = (0.30, 0.70)$) opening at $e_{B,1} = 2$. Completing A's MS1 first realises the large early inflow one period earlier, strictly dominating the reverse ordering under $\gamma < 1$.

Parameters.

$n, s_i, f_i, D^{\text{plan}}$	2, 1, 6, 6
$\text{BAC}_i, \pi_i, CP_i$	60, 0.50, 90
ϕ_A, ϕ_B	(0.70, 0.30), (0.30, 0.70)
$e_{A,1}, e_{A,2}$	1, 4
$e_{B,1}, e_{B,2}$	2, 4
$\bar{\eta}, \Omega_i, \mu_i, \tau_i^{\text{tol}}$	1.0, 2, 0.50, 2

$$Z_{\text{A-first}} = \gamma^0(63 - 30) + \gamma^1(27 - 30) + \gamma^3(90 - 60) = 33 - 2.85 + 25.72 = \mathbf{55.87}.$$

$$Z_{\text{B-first}} = \gamma^0(27 - 30) + \gamma^1(63 - 30) + \gamma^3(90 - 60) = -3 + 31.35 + 25.72 = 54.07 \quad (\text{dominated}).$$

Assertions for pytest.

- A1. MS1_A certified at $t = 1$; MS1_B certified at $t = 2$.
- A2. $Z^* \geq 55.5$.
- A3. $Z^* > 54.07$ (B-first strictly dominated).
- A4. Both projects completed.

C.3 MP-K': MP-K with Constrained B_1

Economic claim. With $B_1 = 20$ (sufficient for only one MS1 spend of 15 at $t = 1$), the MILP allocates entirely to A at $t = 1$, starves B until A's MS1 inflow arrives, then funds B's MS1 at $t = 2$. Partial allocation to B at $t = 1$ earns no milestone inflow (progress threshold not met and $e_{B,1} = 2$ not yet open) and wastes discounted cash.

Parameters. Same as MP-K except $B_1 = 20$.

$$Z^* = \gamma^0(31.5 - 15) + \gamma^1(13.5 - 15) + \gamma^3(13.5 + 31.5 - 30) = 16.5 - 1.425 + 12.861 = \mathbf{27.936}.$$

Note: Spending any amount on B at $t = 1$ (before $e_{B,1} = 2$ opens) accumulates progress but cannot trigger a payment; the discounted outflow is pure waste. The optimal policy is $x_{B,1} = 0$, confirmed by $Z_{\text{partial-B-first}} = 27.686 < 27.936$.

Cash balance trace. $B_1 = 20 \rightarrow 20 - 15 + 31.5 = 36.5 \rightarrow 36.5 - 15 + 13.5 = 35.0 \rightarrow \dots \rightarrow$ final balance positive throughout.

Assertions for pytest.

- A1. $x_{B,1} = 0$ (B starved at $t = 1$).
- A2. $B_t \geq 0$ for all t .
- A3. MS1_A at $t = 1$; MS1_B at $t = 2$.
- A4. $Z^* \geq 27.0$.

C.4 SP-L: Survival Spend Optimal (Single Project)

Economic claim. With $\eta = 0.70 < 1$, each unit spent yields only 0.70 units of progress. $\text{CPI} = 0.70 < 1$ makes $\text{EAC} = \text{BAC}/0.70 = 142.9 > (1 + \mu)\text{BAC}$, so Condition 2 fires permanently. Spending at rate $\bar{x} = \text{BAC}/(\eta \cdot D^{\text{plan}}) = 17.86$ per period keeps $\text{BCWP}_t = \text{BCWS}_t$ ($\text{SPI} = 1$), lapsing Condition 1, so only one condition fires and the counter is frozen. This survival spend is economically optimal: the discounted milestone inflow at $t = 6$ exceeds the total survival cost.

Parameters.

$\text{BAC}_1, \pi_1, CP_1$	100, 0.70, 170
η_1	0.70 (constant)
M_1, θ_1, ϕ_1	1, 1.0, 1.0
$e_{1,1}, D^{\text{plan}}$	6, 8
$\mu_1, \tau_1^{\text{tol}}, \Omega_1$	0.10, 2, 1
$\alpha_1 = 0, \rho_1 = 0$	

Survival allocation. Spend $\bar{x} = 17.857$ per period for $t = 1, \dots, 5$ ($\text{SPI} = 1$ each period, Condition 1 lapsed, counter frozen). At $t = 6$: spend $x_{1,6} = 100/0.70 - 5 \times 17.857 = 53.571$ to reach $P = 1.0$. MS1 certifies. $R_{1,1}^{\text{net}} = 170$ (no advance, no retention).

$$Z^* = \sum_{t=1}^5 \gamma^{t-1} (-17.857) + \gamma^5 (170 - 53.571) = -68.60 + 77.90 = \mathbf{9.298}.$$

Zero-spend baseline: both conditions fire from $t = 1$; counter $2 \rightarrow 1 \rightarrow 0$; terminate $t = 3$. $Z = 0$.

Assertions for pytest.

- A1. $x_{1,t} = 17.857 \pm 0.5$ for $t = 1, \dots, 5$.
- A2. $\text{SPI}_{1,t} = 1.0 \pm 0.02$ for $t = 1, \dots, 5$.
- A3. $\tau_{1,t}^{\text{rem}} = 2$ for all $t \leq 5$ (counter frozen).

A4. MS1 certified at $t = 6$.

A5. $|Z^* - 9.298| \leq 1.0$.

C.5 SP-L': Survival Spend Not Optimal (Single Project)

Economic claim. Same parameters as SP-L except $CP_1 = 80$ ($R_{1,1}^{\text{net}} = 80$). The discounted inflow at $t = 6$ no longer covers the survival cost. Terminating immediately ($Z = 0$) dominates survival.

$$Z_{\text{survive}} = \sum_{t=1}^5 \gamma^{t-1}(-17.857) + \gamma^5(80 - 53.571) = -68.60 + 18.84 = -49.76 < 0.$$

Assertions for pytest.

A1. $\sum_t x_{1,t} \leq 1.0$ (MILP allocates essentially nothing).

A2. Project terminates at $t \leq 3$.

A3. $Z^* = 0 \pm 0.5$.

C.6 SP-M: Survival Spend Optimal (Multi-Project)

Economic claim. A Good project (high η , certifies MS1 immediately) generates an early inflow that funds the survival allocations for a Bad project (low η , must survive to $t = 6$ for its milestone). The portfolio value exceeds completing Good alone.

Parameters.

Good: BAC = 60, $\eta = 1$, $M = 1$, $\theta = 1.0$, $e = 1$, $CP = 90$	$R^{\text{net}} = 90$, $x_1 = 60$
Bad: same as SP-L above (BAC = 100, $\eta = 0.70$, $CP = 170$)	
$B_1 = 90$, $\gamma = 0.95$	

$$Z^* = \underbrace{(90 - 60)}_{Z_{\text{Good}}} + \underbrace{9.298}_{Z_{\text{Bad}}} = 30 + 9.298 = \mathbf{39.298}.$$

Cash balance stays non-negative throughout (verified: $B_t \geq 30$ for all t).

Assertions for pytest.

A1. Good completed at $t = 1$.

A2. Bad MS1 certified at $t = 6$.

A3. $B_t \geq 0$ for all t .

A4. $|Z^* - 39.298| \leq 1.5$.

C.7 SP-M': Survival Spend Not Optimal (Multi-Project)

Economic claim. Same as SP-M but Bad's $CP = 80$ (as in SP-L'). The MILP funds only Good and immediately terminates Bad. $Z^* = Z_{\text{Good}} = 30$.

Assertions for pytest.

- A1. Good completed.
- A2. Bad terminated at $t \leq 3$.
- A3. $|Z^* - 30.0| \leq 1.0$.

D Group 3: Advance Payment and Recovery Mechanics

What is being proven: advance arrives at $t = s_i - 1$ (period 0 for single-project cases with $s_i = 1$), recovery deductions apply once ψ_i is crossed, deductions stop once advance is fully recovered, and the payment identity $A_i + \sum_j R_{i,j}^{\text{net}} + R_i^{\text{ret}} = CP_i$ holds exactly.

D.1 SP-2: Advance Harvest, Efficiency Decay, Forced Termination

Economic claim. A large advance combined with rapidly decaying efficiency ($\eta_1 = 1.0$, $\eta_{t \geq 2} = 0.05$) makes completion economically impossible. The MILP takes the advance at $t = 0$ (i.e. $s_1 - 1 = 0$) and accepts termination. With $\gamma < 1$, the advance inflow at $t = 0$ is worth more than the discounted settlement at $t = 2$, giving $Z^* = 2.925 > 0$.

Parameters.

BAC_1, π_1, CP_1	100, 0, 100
α_1, A_1	0.30, 30
$\delta^{\text{rec}}, \psi, \rho_1$	0.25, 0.10, 0
$\Omega_1, \mu_1, \tau_1^{\text{tol}}$	1, 0.05, 1
$\bar{\eta}$	1.0 at $t = 1$; 0.05 for $t \geq 2$
H, B_1	6, 300

Zero spend: counter $1 \rightarrow 0$ at $t = 1$, termination at $t = 2$. $P = 0$. $R^{\text{term}} = -30$.

$$Z^* = \underbrace{\gamma^0 \cdot 30}_{\text{advance } t=0} + \underbrace{\gamma^2 \cdot (-30)}_{\text{settlement } t=2} = 30 - 27.075 = \mathbf{2.925}.$$

Assertions for pytest.

- A1. $\sum_t x_{1,t} \leq 1.0$.
- A2. Project terminates at $t \leq 3$.
- A3. $R_1^{\text{term}} = -30 \pm 1$.
- A4. $|Z^* - 2.925| \leq 0.5$.

D.2 SP-2a: Preferred Allocation Before Termination

Economic claim. Adding a single milestone ($M = 1$, $\theta = 1.0$, $e = 1$) with sufficient CP makes it strictly better to spend 100 at $t = 1$ (certifying and completing the project) than to accept termination.

Parameters.

$\text{BAC}_1, \pi_1, CP_1$	100, 0.20, 120
$\alpha_1, A_1, \delta^{\text{rec}}, \psi$	0.30, 30, 0.25, 0.10
$\eta_1, M_1, \theta_1, e_{1,1}$	1.0, 1, 1.0, 1
$\mu_1, \tau_1^{\text{tol}}$	0.30, 1

Advance at $t = 0$: +30. Spend 100 at $t = 1$: $P = 1.0$, MS1 certifies. Recovery: $\delta^{\text{rec}} \cdot CP = 0.25 \times 120 = 30 = A$. Fully recovered. $R_{1,1}^{\text{net}} = 120 - 30 = 90$.

$$Z_{\text{spend}} = 30 + \gamma^1(90 - 100) = 30 - 9.5 = \mathbf{20.5}.$$

$$Z_{\text{zero}} = 30 + \gamma^2(-30) = 30 - 27.075 = 2.925.$$

Spending strictly preferred: $20.5 > 2.925$.

Assertions for pytest.

A1. $x_{1,1} = 100 \pm 0.5$.

A2. MS1 certified at $t = 1$.

A3. $|Z^* - 20.5| \leq 1.0$.

A4. Status: `completed`.

D.3 SP-2b: No Preferred Allocation

Economic claim. Same setup as SP-2a but $CP_1 = 60$ (small contract value). The milestone payment after advance recovery ($R_{1,1}^{\text{net}} = 60 - 30 = 30$) does not cover the spend of 100. Terminating is optimal.

Parameters. As SP-2a except $CP_1 = 60, \pi_1 = 0$.

$$Z_{\text{spend}} = 30 + \gamma^1(30 - 100) = 30 - 66.5 = -36.5.$$

$$Z_{\text{zero}} = 30 + \gamma^2(-30) = 2.925.$$

Zero spend preferred: $2.925 > -36.5$.

Assertions for pytest.

A1. $\sum_t x_{1,t} \leq 1.0$.

A2. Project terminates at $t \leq 3$.

A3. $|Z^* - 2.925| \leq 0.5$.

D.4 SP-2c: Increased Tolerance, Inevitable Termination

Economic claim. Raising $\tau_1^{\text{tol}} = 3$ with the same efficiency decay delays termination to $t = 4$ but does not enable completion. Higher discounting on the negative settlement payment increases Z^* .

Parameters. As SP-2 except $\tau_1^{\text{tol}} = 3$. Counter: $3 \rightarrow 2 \rightarrow 1 \rightarrow 0$ at $t = 1, 2, 3$. Termination at $t = 4$.

$$Z^* = 30 + \gamma^4(-30) = 30 - 24.435 = \mathbf{5.565}.$$

Compare SP-2 baseline ($\tau = 1$): $Z^* = 2.925$. Higher τ is better for the contractor even when termination is inevitable.

Assertions for pytest.

A1. $\tau_{1,t}^{\text{rem}} \in \{2, 1, 0\}$ at $t = 1, 2, 3$.

A2. Termination at $t = 4$.

A3. $R_1^{\text{term}} = -30 \pm 1$.

A4. $|Z^* - 5.565| \leq 0.5$.

D.5 SP-2d: Increased Tolerance, Efficiency Recovery Enables Completion

Economic claim. With $\tau_1^{\text{tol}} = 4$ and an efficiency schedule that recovers to $\eta = 1.0$ at $t = 4$, the project survives the low-efficiency periods and completes. Certifying at $t = 4$ gives $Z^* = 21.855$, strictly better than termination at $t = 5$ ($Z = 6.787$).

Parameters. As SP-2a except $\tau_1^{\text{tol}} = 4$, $e_{1,1} = 4$, and $\bar{\eta} = 0.05$ for $t = 1, 2, 3$ then $\bar{\eta} = 1.0$ for $t \geq 4$.

Zero spend at $t = 1, 2, 3$: counter $4 \rightarrow 3 \rightarrow 2 \rightarrow 1$. At $t = 4$: spend 100, $P = 1.0$, MS1 certifies. Counter stays at 1 (neither condition fires at $t = 4$).

$$Z_{\text{certify}} = 30 + \gamma^4(90 - 100) = 30 - 8.145 = \mathbf{21.855}.$$

$$Z_{\text{zero}} = 30 + \gamma^5(-30) = 30 - 23.114 = 6.787.$$

Assertions for pytest.

A1. $\tau_{1,t}^{\text{rem}} \in \{3, 2, 1\}$ at $t = 1, 2, 3$.

A2. $x_{1,4} = 100 \pm 0.5$.

A3. MS1 certified at $t = 4$.

A4. $|Z^* - 21.855| \leq 1.0$.

D.6 SP-A: Late Recovery Threshold ($\psi = 0.40$)

Economic claim. With $\psi = 0.40$ and equal milestone weights $\phi_j = 1/3$, cumulative certified payments first exceed ψ at MS2 ($\text{cum_}\phi = 2/3 \geq 0.40$). MS1 payment is therefore full (no deduction); MS2 and MS3 have deductions.

Parameters.

BAC_1, π_1, CP_1	120, 0.50, 180
$\alpha_1, A_1, \psi, \delta^{\text{rec}}$	0.20, 36, 0.40, 0.30
$e_{1,1}, e_{1,2}, e_{1,3}$	1, 3, 5

$$R_{1,1}^{\text{net}} = 60, \quad R_{1,2}^{\text{net}} = 42, \quad R_{1,3}^{\text{net}} = 42.$$

$$Z^* = \gamma^0(36 + 60 - 40) + \gamma^2(42 - 40) + \gamma^4(42 - 40) = 56 + 1.805 + 1.629 = \mathbf{59.43}.$$

Assertions for pytest.

A1. $R_{1,1}^{\text{net}} = 60 \pm 0.5$ (no deduction, ψ not yet crossed).

A2. $R_{1,2}^{\text{net}} = 42 \pm 0.5$.

A3. $R_{1,3}^{\text{net}} = 42 \pm 0.5$.

A4. $|Z^* - 59.43| \leq 1.5$.

D.7 SP-B: Advance Fully Recovered Before Final Milestone

Parameters.

BAC_1, π_1, CP_1	60, 0.50, 90
$\alpha_1, A_1, \psi, \delta^{\text{rec}}, \rho_1$	0.20, 18, 0.10, 0.30, 0
$e_{1,1}, e_{1,2}, e_{1,3}$	1, 2, 3

$$R_{1,1}^{\text{net}} = 21, \quad R_{1,2}^{\text{net}} = 21, \quad R_{1,3}^{\text{net}} = 30.$$

$$Z^* = \gamma^0(18 + 21 - 20) + \gamma^1(21 - 20) + \gamma^2(30 - 20) = 19 + 0.95 + 9.025 = \mathbf{28.975}.$$

Assertions for pytest.

A1. $R_{1,3}^{\text{net}} = 30 \pm 0.5$ (advance already recovered; no deduction on MS3).

A2. $|Z^* - 28.975| \leq 1.0$.

E Group 4: Retention

E.1 SP-C: Retention Only, Lump Release at Completion

Parameters. $BAC_1 = 60, \pi_1 = 0.50, CP_1 = 90, M_1 = 3, \rho_1 = 0.10, \alpha_1 = 0. e_{1,1} = 1, e_{1,2} = 3, e_{1,3} = 5. R_{1,j}^{\text{net}} = 27$ each; $R_1^{\text{ret}} = 9$ at $t = 5$.

$$Z^* = \gamma^0(27 - 20) + \gamma^2(27 - 20) + \gamma^4(36 - 20) = 7 + 6.318 + 13.032 = \mathbf{26.35}.$$

Assertions for pytest.

A1. $R_{1,j}^{\text{net}} = 27 \pm 0.5$ for $j = 1, 2, 3$.

A2. $R_1^{\text{ret}} = 9 \pm 0.5$, released only at $t = 5$.

A3. Retention release event is at the final milestone period — not before.

A4. $|Z^* - 26.35| \leq 1.0$.

E.2 SP-D: Retention and Advance Recovery Simultaneously Active

Parameters. $BAC_1 = 60$, $\pi_1 = 0.50$, $CP_1 = 90$, $\rho_1 = 0.10$, $\alpha_1 = 0.20$, $A_1 = 18$, $\psi = 0.10$, $\delta^{\text{rec}} = 0.30$. $e_{1,1} = 1$, $e_{1,2} = 3$, $e_{1,3} = 5$. $R_{1,1}^{\text{net}} = 18$, $R_{1,2}^{\text{net}} = 18$, $R_{1,3}^{\text{net}} = 27$; $R_1^{\text{ret}} = 9$.

$$Z^* = \gamma^0(18 + 18 - 20) + \gamma^2(18 - 20) + \gamma^4(27 + 9 - 20) = 16 - 1.805 + 13.032 = \mathbf{27.227}.$$

Assertions for pytest.

- A1. $R_{1,1}^{\text{net}} = 18 \pm 0.5$, $R_{1,2}^{\text{net}} = 18 \pm 0.5$.
- A2. $R_{1,3}^{\text{net}} = 27 \pm 0.5$ (retention only on MS3).
- A3. $R_1^{\text{ret}} = 9 \pm 0.5$ at $t = 5$.
- A4. Payment identity: $A_1 + R_{1,1}^{\text{net}} + R_{1,2}^{\text{net}} + R_{1,3}^{\text{net}} + R_1^{\text{ret}} = 90 = CP_1$.
- A5. $|Z^* - 27.227| \leq 1.0$.

F Group 5: Termination Conditions in Isolation

Note: The cure-period counter decrements only when *both* conditions fire simultaneously. When only one fires, the counter stays at τ^{tol} .

F.1 SP-E: Only Condition 1 Fires

Assertions for pytest.

- A1. $SPI_{1,t} < 1$ for $t \geq 3$.
- A2. $CPI_{1,t} = 1.0 \pm 0.01$ for all t .
- A3. $\tau_{1,t}^{\text{rem}} = 3$ for all t .
- A4. Status never `terminated`.

F.2 SP-F: Only Condition 2 Fires

Assertions for pytest.

- A1. $SPI_{1,t} > 1$ for $t \leq 10$.
- A2. $EAC_{1,t} > 1.05 \cdot BAC$ from $t = 1$.
- A3. $\tau_{1,t}^{\text{rem}} = 3$ for all observed periods.
- A4. Status never `terminated`.

F.3 SP-G: Both Fire, Condition 1 Lapses, Counter Resets

Efficiency schedule: $\eta_1 = \eta_2 = 0.3$; $\eta_3 = 1.0$; $\eta_{t \geq 4} = 0.3$.

Assertions for pytest.

- A1. $\tau_{1,1}^{\text{rem}} = 3$ (only Cond. 2 at $t = 1$).
- A2. $\tau_{1,2}^{\text{rem}} = 2$ (both conditions; decrement).
- A3. $\tau_{1,3}^{\text{rem}} = 3$ (Cond. 1 lapses after spike; reset).
- A4. Status at $t = 3$: **active**.

G Group 6: Cure Period and Termination Execution

G.1 SP-5: Zero Efficiency, Exact Counter Boundary

Parameters: $\eta = 0$, $\tau^{\text{tol}} = 3$, $A_1 = 20$. Advance at $t = 0$.

$$Z^* = \gamma^0 \cdot 20 + \gamma^3 \cdot (-20) = 20 - 17.15 = \mathbf{2.85}.$$

Assertions for pytest.

- A1. $\tau_{1,t}^{\text{rem}} \in \{2, 1, 0\}$ at $t = 1, 2, 3$.
- A2. Termination at $t = 4$.
- A3. $R_1^{\text{term}} = -20 \pm 0.5$.
- A4. $|Z^* - 2.85| \leq 0.5$.

G.2 SP-H: Partial Progress at Termination, Positive Settlement

Parameters: $\eta = 0.5$, $A_1 = 10$, $\tau^{\text{tol}} = 2$, $\theta_1 = 0.30$, $e_{1,1} = 1$. Advance at $t = 0$. Spend 60 at $t = 1$, MS1 certifies ($R = 30$). Termination at $t = 3$. $R^{\text{term}} = 30$.

$$Z^* = \gamma^0(10 + 30 - 60) + \gamma^1(0) + \gamma^2(30) = -20 + 0 + 27.075 = \mathbf{7.075}.$$

Assertions for pytest.

- A1. MS1 certified at $t = 1$.
- A2. $R_1^{\text{term}} = 30 \pm 0.5$.
- A3. Termination at $t = 3$.
- A4. $|Z^* - 7.075| \leq 1.0$.

G.3 SP-I: Cure Period Interrupted and Restarted

Counter trajectory: $\tau_t^{\text{rem}} = [3, 2, 3, 2, 1, 0]$ at $t = 1 \dots 6$; termination at $t = 7$.

Assertions for pytest.

- A1. $\tau_{1,2}^{\text{rem}} = 2$.
- A2. $\tau_{1,3}^{\text{rem}} = 3$ (reset after spike).
- A3. $\tau_{1,6}^{\text{rem}} = 0$.
- A4. Termination at $t = 7$ (not $t = 4$).

H Group 7: Portfolio Cash Balance and Coupling

H.1 MP-3: Advance Harvest Funds Concurrent Good Project

$$Z^* = \gamma^0(30 + 60 - 40) + \gamma^1(60 - 40) + \gamma^2(-30) = 50 + 19 - 27.075 = \mathbf{41.925}.$$

Assertions for pytest.

- A1. $B_t \geq 0$ for all t .
- A2. Good completed.
- A3. $|Z^* - 41.925| \leq 1.5$.

H.2 MP-A: Sequential Funding

With $B_1 = 30$, only one project can be funded at $t = 1$.

Assertions for pytest.

- A1. $x_{2,1} = 0$ (P2 starved at $t = 1$).
- A2. $B_t \geq 0$ for all t .
- A3. Both completed.
- A4. $Z^* \geq 50$.

H.3 MP-B: Negative Settlement Drains Shared Balance

$$Z^* = \gamma^0(18 - 30 + 45) + \gamma^1(45 - 30) + \gamma^2(-18) = 33 + 14.25 - 16.245 = \mathbf{31.005}.$$

Assertions for pytest.

- A1. $R_{\text{Term}}^{\text{term}} = -18 \pm 0.5$.
- A2. $|Z^* - 31.005| \leq 1.5$.

I Group 8: Portfolio Selection Under Constraints

I.1 MP-2: Good vs. Bad, Unconstrained Cash

$$Z_G = \gamma^0(60 - 40) + \gamma^4(60 - 40) = 20 + 16.29 = 36.29, \quad Z_B = 1.5, \quad Z^* = \mathbf{37.79}.$$

Assertions for pytest.

- A1. $\sum_t x_{\text{Bad},t} \leq 1.0$.
- A2. Bad terminates at $t \leq 4$.
- A3. Good completed.
- A4. $|Z^* - 37.79| \leq 1.5$.

I.2 MP-4: Three Projects, Low Always Terminates

$$Z^* = \gamma^0(45 + 36 - 60) + \gamma^1(45 + 36 - 60) + 0 = 21 + 19.95 = \mathbf{40.95}.$$

Assertions for pytest.

A1. High and Mid completed.

A2. Low terminated.

A3. $|Z^* - 40.95| \leq 1.5$.

A4. $B_t \geq 0$.

I.3 MP-C: Cash Sufficient for Only One Project

$$Z^* = \gamma^0(45 - 30) + \gamma^2(45 - 30) = 15 + 13.54 = \mathbf{28.54}.$$

Assertions for pytest.

A1. Lo terminated; Hi completed.

A2. $|Z^* - 28.54| \leq 1.5$.

A3. $B_t \geq 0$.

I.4 MP-D: Bridge Cash Unlocks Major Projects

Assertions for pytest.

A1. Bridge completed.

A2. $Z^* > 28$.

A3. $B_t \geq 0$.

J Group 9: Horizon and Staggered Starts

J.1 MP-1: Two Non-Overlapping Projects

$$Z_A = \gamma^0 \times 10 + \gamma^2 \times 10 = 19.025, \quad Z_B = \gamma^4 \times 10 + \gamma^7 \times 10 = 8.145 + 6.983 = 15.128.$$

$$Z^* = 19.025 + 15.128 = \mathbf{34.153}.$$

Assertions for pytest.

A1. $H = 10$.

A2. $x_{A,t} = 0$ for $t \geq 5$; $x_{B,t} = 0$ for $t \leq 4$.

A3. $|Z^* - 34.153| \leq 1.5$.

J.2 MP-E: Three Projects with Staggered Starts

$$Z_1 = 28.54, \quad Z_2 = 23.83, \quad Z_3 = 17.24, \quad Z^* = \mathbf{69.61}.$$

Assertions for pytest.

A1. $H = 14$.

A2. $x_{1,t} = 0$ for $t \geq 5$; $x_{2,t} = 0$ for $t \leq 3$ and $t \geq 9$; $x_{3,t} = 0$ for $t \leq 8$.

A3. All three completed.

A4. $|Z^* - 69.61| \leq 2.0$.

K Summary Table

Case	Group	Pattern tested	Outcome	Z^*	Key assertion
SP-1	G1	$\eta = 1$, e pins timing	Completed	19.03	Spend $t = 1, 3$; MS2 blocked until $e = 3$
SP-3	G1	3 MS, spend at each	Completed	27.17	Spend $t = 1, 3, 5$
SP-J	G1	$e_{i,j}$ Progress jump; e blocks MS2	Completed	0	MS2 not certified until $t = 3$
SP-4	G2	Defer spend to $e_{i,2} = 5$	Completed	36.29	$Z^* > 29.5$ (front-load dominated)
MP-K	G2	A-first beats B-first	Both comp.	55.87	$Z > 54.07$; A's large MS1 first
MP-K'	G2	Constrained B_1 , A-first	Both comp.	27.94	$x_{B,1} = 0$; defer B to $t = 2$
SP-L	G2	Survival spend optimal (SP)	Completed	9.30	SPI= 1 freezes counter; $Z > 0$
SP-L'	G2	Survival not optimal (SP)	Terminated	0	$Z_{\text{survive}} < 0$
SP-M	G2	Survival optimal (MP)	Both comp.	39.30	Good funds Bad survival
SP-M'	G2	Survival not optimal (MP)	Good comp.	30.00	Bad terminated, Good only
SP-2	G3	Advance harvest; γ gain	Terminated	2.925	Advance $t = 0$; $Z > 0$ via discounting
SP-2a	G3	Preferred alloc, certify	Completed	20.50	Spend 100 beats terminate
SP-2b	G3	No preferred alloc	Terminated	2.925	Zero spend wins; CP too small
SP-2c	G3	$\tau = 3$, inevitable term.	Terminated	5.565	Later settle; higher Z via γ
SP-2d	G3	$\tau = 4$, η recovers	Completed	21.855	Certify at $t = 4$ beats terminate
SP-A	G3	Late ψ ; MS1 full	Completed	59.43	MS1 full; MS2,3 deducted
SP-B	G3	Recovery done before MS3	Completed	28.975	MS3 full; $R = 30$
SP-C	G4	Retention; lump release	Completed	26.35	Release $t = 5$; $R^{\text{ret}} = 9$

Case	Group	Pattern tested	Outcome	Z^*	Key assertion
SP-D	G4	Retention + advance	Completed	27.23	MS3 retention only
SP-E	G5	Only Cond. 1; counter fixed	Active	—	$\tau = 3$ always
SP-F	G5	Only Cond. 2; counter fixed	Active	—	$\tau = 3$ always
SP-G	G5	Reset mid-cure	Active	—	$\tau = [3, 2, 3]$
SP-5	G6	$\eta = 0$; counter boundary	Terminated	2.85	$\tau = [2, 1, 0]$; $Z > 0$ via γ
SP-H	G6	Partial progress; $R^{\text{term}} > 0$	Terminated	7.075	$R^{\text{term}} = 30$; spend preferred
SP-I	G6	Cure reset; late termination	Terminated	—	Term $t = 7$ not $t = 4$
MP-3	G7	Advance funds concurrent	Mixed	41.93	$B_t \geq 0$; discounted Z
MP-A	G7	Sequential funding	Both comp.	≥ 50	$x_{2,1} = 0$
MP-B	G7	Negative settlement drains B	Mixed	31.01	Settlement -18 visible
MP-2	G8	Good vs. Bad	Bad term.	37.79	Good completes; Bad harvested
MP-4	G8	Low always terminates	Low term.	40.95	H&M complete; Low terminates
MP-C	G8	One project's worth of cash	Lo term.	28.54	Hi completes; Lo starved
MP-D	G8	Bridge unlocks majors	All comp.	> 28	Option value realised
MP-1	G9	Non-overlapping; additive Z	Both comp.	34.15	$Z = Z_A + Z_B$ discounted
MP-E	G9	Staggered starts; $H = 14$	All comp.	69.61	Spend at each $e_{i,j}$; windows respected

Table 3: All test cases with $\gamma = 0.95$ and earliest certification dates. Z^* values include the new advance-at- t_0 convention. Cases marked — have no meaningful Z^* (focus is counter behaviour).